OMT FOR BOARD REVIEW

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OBJECTIVES

- Define and differentiate active and passive, direct and indirect techniques
- Apply these definitions to different techniques used in the osteopathic profession
- Describe the procedure for various OMT modalities
- Describe the physiologic principles of the Golgi tendon and Muscle spindle as they relate to muscle energy
- Identify contraindications for these techniques
- Review Somatic Dysfunction and Fryette's Principles of vertebral motion
- Review other high-Yield Board Material

PASSIVE VS. ACTIVE TECHNIQUES

Passive: Techniques that are performed by the physician without any active participation by the patient (Example: Counterstrain).

Active: Techniques that require significant participation by the patient, guided by the physician. Voluntary muscle contraction and respiratory effort are two examples of active participation.

DiGiovanna, et. al. page 94

PASSIVE VS. ACTIVE

Passive:

HVLA
Counterstrain
Soft Tissue
MFR
BLT
OCMM
FPR
Still's Technique
Lymphatics
Visceral

Active:

Muscle Energy
MFR
Lymphatics
BLT
Visceral

DIRECT VS. INDIRECT

- **Direct:** Direct techniques are ones in which the restricted joint or tissue is initially taken in the direction of the restriction to motion. The restricted part is carried to the barrier at the beginning of the treatment. In some, an operator-generated force allows the joint or tissue to move beyond the restrictive barrier to motion. In others, the motion occurs gradually during the treatment.
- **Indirect:** Indirect techniques are those that initially position the joint or tissue away from a barrier to motion and toward the relative ease or freedom of motion. Indirect techniques allow neural mechanisms or fascial tensions to be altered so as to permit improved motion of the joint or tissue.
- **Combination:** There are a few techniques in which part of the technique is indirect and then a direct component is added or, conversely, a technique may begin as a direct technique then an indirect component

DiGiovanna, et. al. page 94

DIRECT VS. INDIRECT

Direct:

HVLA
Muscle Energy
MFR
Still Technique (second part)
Articulatory/Springing
Soft Tissue
OCMM
Lymphatics
Visceral

Indirect:

Counterstrain
MFR
BLT
Still Technique (first part)
Facilitated Positional Release
OCMM
Lymphatics
Visceral

FRYETTE'S PRINCIPLES OF VERTEBRAL MOTION

1st Principle (Type I):	Group Curves: Involving 2 or more segments. Side bending and rotation occur to opposite sides. Neutral (no change with flexion and/or extension).
2nd Principle (Type II):	Individual segment (often pathologic). Non-neutral (flexion or extension). Rotation occurs in the same direction and side bending. Can occur at apex or either end of group curve.
3rd Principle:	Motion in one plane effects motion in all other planes.

SOMATIC DYSFUNCTION

T.A.R.T.

Always name for the relative FREEDOM of motion

Vertebra are named as a functional vertebral unit (two vertebra and intervening disc- ie. C3 motion on C4)

Vertebral unit

NAMING SOMATIC DYSFUNCTION

Example:

T4 Transverse Process posterior on the right

Transverse Processes feel equal when patient flexes head down to T4 level

Right transverse process feels more posterior when head is extended to T4 level

Diagnosis??

Fryette's type I or type II??

What would be the starting position with direct technique??

What would be the starting position with indirect technique??

MUSCLE ENERGY

Muscle Energy:

Active and Direct.

The directed patient action is from a precisely controlled position against a defined resistance by the osteopathic practitioner.*

Developed by Fred L. Mitchell, Sr. D.O.

Isometric, Isotonic and Isolytic Contractions

Golgi tendon Reflex.

MUSCLE ENERGY PHYSIOLOGY- GOLGI TENDON REFLEX

Located in mammalian tendons between the muscle and the tendon insertion.

One tendon organ for every 3-15 muscle fibers.

Directly measure the degree of muscle tension and convey this information to the CNS.

In series with extrafusal fibers and are stretched when muscle contracts.

Increased tension-->muscle relaxation.

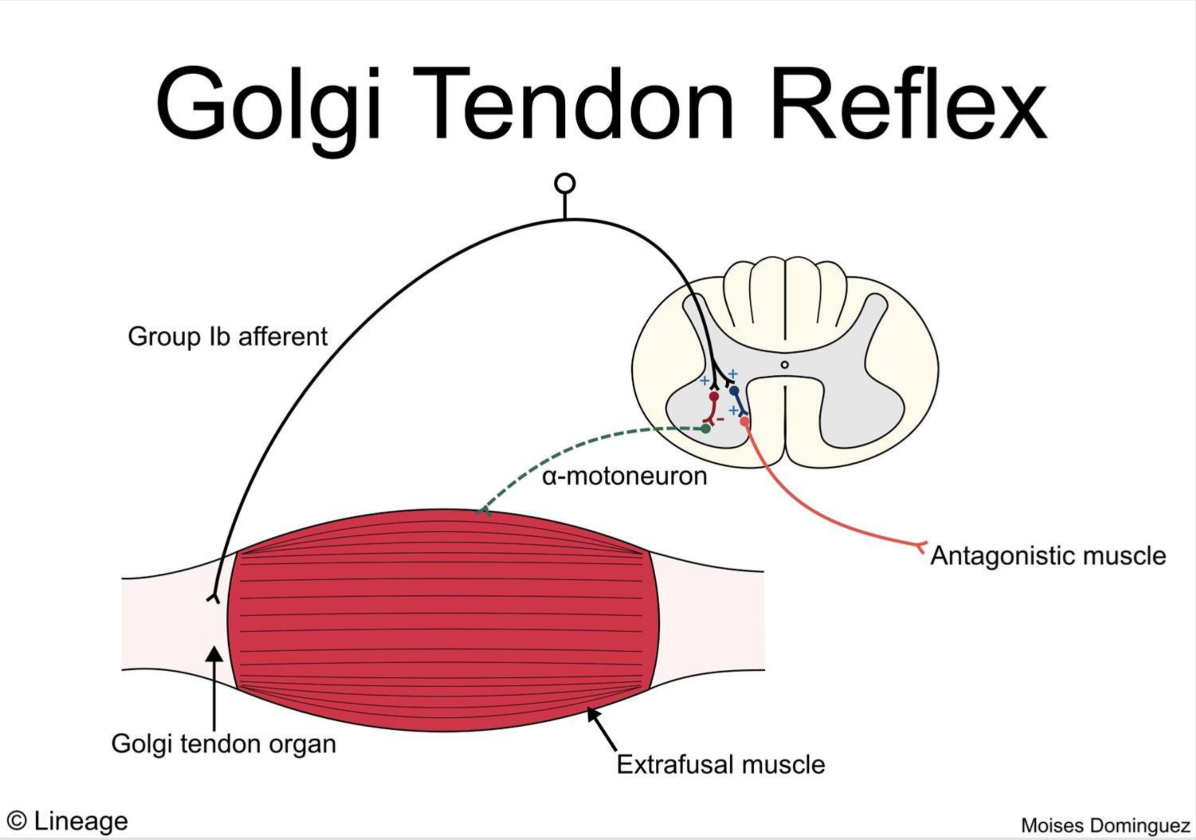
MUSCLE ENERGY PHYSIOLOGY- GOLGI TENDON REFLEX

Afferent Neurons:

Ib Group: Myelinated.

Synapse with inhibitory interneurons (and interneurons that ascend to higher CNS).

Inhibitory interneurons synapse with large alpha motor neurons.



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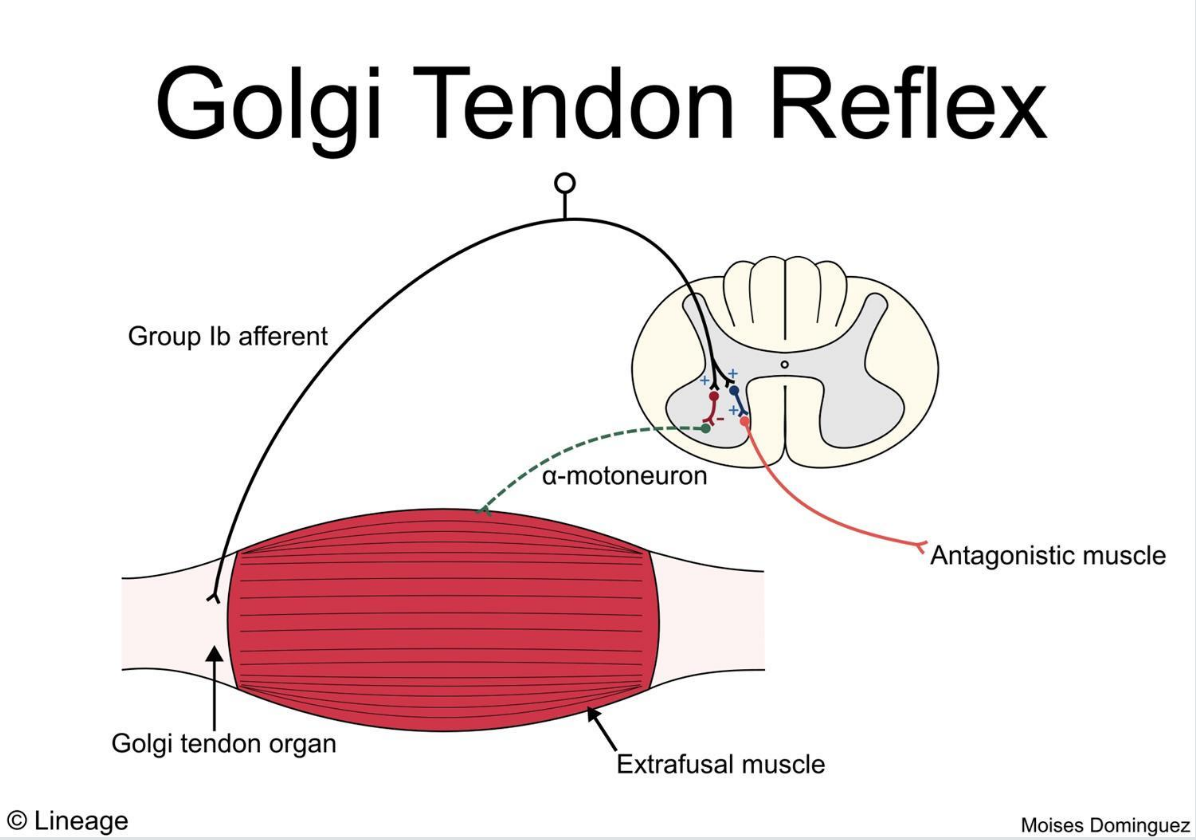
MUSCLE ENERGY PHYSIOLOGY- GOLGI TENDON REFLEX

Efferent Neurons:

Alpha motor neurons:

Inhibited by inhibitory interneurons.

Results in reflex relaxation of the muscle (same muscle that the Ib afferents originate).



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MUSCLE ENERGY PROCEDURE

1. **Diagnose** S.D.
2. Engage the dysfunctional **(restrictive)** BARRIER.
3. Patient contracts (**appropriate level of force to engage level**) against physicians resistance for 3-4 seconds.
4. Patient relaxes **(complete relaxation)** for 2 seconds.
5. Physician engages new barrier (**feather's edge**).
6. Repeat the sequence as appropriate (**3-5 times**).
7. **Reassess**

Video: **Hamstring Muscle Energy**

MUSCLE ENERGY COMMON MISTAKES

1. Wrong diagnosis
2. Physician doesn’t accurately monitor with palpation of involved joint
3. Patients used too much force
4. Muscle contraction is too short
5. The patient is not allowed to totally relax before repositioning
6. Physician forgets to reassess

MUSCLE ENERGY INDICATIONS

Treatment of:

Motion restriction of an individual joint (Type II SD) or an abnormally short monoarticular muscle.

Motion restriction of more than one adjacent joint (Type I SD) or an abnormally short polyarticular muscle

Motion restriction due to fibrosis and adhesions (Isolytic)

Articular hypermobility due to muscular weakness (Isokinetic)

Strengthen a weakened muscle or group of muscles

Reduce edema or passive congestion

MUSCLE ENERGY CONTRAINDICATIONS

Absolute:

Lack of patient consent or cooperation

Relative:

Infection hematoma or tear of involved muscle (cross extensor reflex may be used)

Fracture or dislocation of involved joint (cross extensor reflex may be used)

Rheumatological condition causing instability of the cervical spine

Undiagnosed joint swelling of the involved joint

Positioning that compromises vasculature

Painful tissue damage to ligamentous, tendon or muscle tissues (cross extensor reflex may be used)

Centrally mediated muscle spasm

Immediate postsurgical period

Low vitality secondary to illness (too much exertion)

Inability to participate due to cooperation/understanding (pediatric patient- torticollis)

Tissue fragility

HVLA DEFINITION

- “An osteopathic technique employing a rapid, therapeutic force of brief duration that travels a short distance within the anatomic range of motion of a joint, and that engages the restrictive barrier in one or more planes of motion to elicit release of restriction.“

- (ECOP 2006)

HVLA

Passive and Direct

Barrier is engaged by physician

Physician applies a sudden, gentle, increase of force of a small distance

"END FEEL" is critical for success (barrier must be appropriately engaged).

Thrusting Principles:

- Relax soft tissues prior.
- Barrier is engaged by physician.
- Maintain position to not lose the "lock out."
- Controlled force.
- Localized.

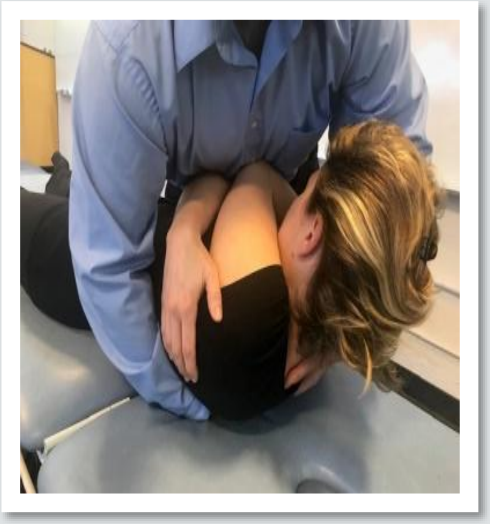
HVLA PROCEDURE

1. Diagnose Somatic Dysfunction
2. Loosen soft tissue before thrust (soft tissue, ME, etc.)
3. Place dysfunctional joint against its restrictive motion barrier(s) and lock out all other motion both above and below the specific joint to be manipulated.
4. Apply activating force, impulse, to restrictive motion barriers while maintaining the locking out of other motions.
5. Hold position briefly after the thrust before return to neutral.
6. Re-evaluate joint motion.

Video: Lumbar HVLA

HVLA INDICATIONS

•<u>Specific</u> joint restriction (somatic dysfunction) with hard barrier; i.e., <u>solid</u> <u>end</u> <u>feel</u>



•Not indicated for use where there is anatomic or pathologic change at the joint (e.g., advanced DJD, contracture)

HVLA ABSOLUTE CONTRAINDICATIONS

Absence of somatic dysfunction

Lack of patient consent and/or cooperation

"Rubbery" quality of soft tissue restriction

Vertigo, change of or loss of consciousness with cervical extension and rotation. (This is an indication of vertebral artery occlusive disease/ vertebral-basilar insufficiency)

Fracture/ dislocation/ spinal instability

Achondroplastic dwarfism

Septic joint, inflammatory joint disease

Vasculitis/ history of CVA

Pathologic instability of the OA joint (i.e. RA, Down Syndrome/Trisomy 21)- especially C1-2

Ehler-Danlos Hypermobility Syndrome

Ankylosing spondylosis

Klippel-Feil syndrome

- Congenital fusion of C2-7

Metabolic bone disease:

- Rickets or Osteomalacia
- Vitamin D deficiency
- Hypophosphatemia
- Hyperparathyroidism
- Corticosteroid overuse
- Paget's disease

HVLA RELATIVE CONTRAINDICATIONS

Acute whiplash / severe muscle spasm / strain/sprain
Herniated disc, acute radiculopathy
Spondylolisthesis
Cancer
Mild to moderate osteoporosis, osteopenia
Neurologic disease (cerebral palsy, Parkinson's, etc)
Pregnancy
Extremes of Age
Toddlers
Geriatric patients
Hypermobility

SOFT TISSUE TECHNIQUES

•"A system of diagnosis and treatment directed toward tissues other than skeletal or arthrodial elements."-

"It is a direct technique that usually involves lateral stretching, linear stretching, deep pressure, traction and/or separation of muscle origin and insertion while monitoring tissue response and motion changes by palpation. Also called myofascial treatment."-

•*Glossary of Osteopathic Terminology*

SOFT TISSUE INDICA

- Relax hypertonic muscles and reduce spasm
- Stretch and increase elasticity of shortened fascial structures
- Enhance circulation to local myofascial structures
- Improve local tissue nutrition, oxygenation, and removal of metabolic wastes
- Improve abnormal somato-somatic and somato-visceral reflex activities, thus improving circulation in areas of the body remote from the area being treated
- Diagnostically to identify areas of restricted motion, tissue texture changes and sensitivity
- Observe tissue response to the application of manipulative technique
- Improve local and systemic immune response
- Provide a general state of relaxation
- Provide a general state of tonic stimulation by stimulating the stretch reflex in hypotonic muscles

Video: Cervical Soft Tissue

SOFT TISSUE CONTRAINDICATIONS

Relative: acutely injured muscles, tendons, ligaments, or joint capsules. ST treatment may do additional damage to these structures or increase the amount of pain the patient experiences.

Absolute: fracture or dislocation, neurologic entrapment syndromes, serious vascular compromise, local malignancy, local infection (cellulitis, abscess, septic arthritis, osteomyelitis), bleeding disorders

The physician may still work proximal or distal to the affected area and may alter the patient's position or technique to achieve some beneficial effect.

MYOFASCIAL RELEASE (DIRECT OR INDIRECT)

Definition:

Myofascial release (MFR)… "is a system of diagnosis and treatment first described by Andrew Taylor Still and his early students, which engages continual palpatory feedback to achieve release of myofascial tissues."

Direct, indirect, or combined treatment

Activating forces include: Inherent (intrinsic) force; Respiratory force (cooperation/assist); Patient cooperation; Physician-guided force; Springing/vibration

MYOFASCIAL RELEASE

Indications:

- SD involving myofascial tissues or other connective tissues

Contraindications:

- Absolute:
  - Lack of patient consent and/or cooperation
- Relative:
  - Caution with
    - fractures
    - open wounds
    - soft tissue or bony infections abscesses
    - Deep venous thrombosis
    - Anticoagulation, disseminated or focal neoplasm
    - recent post-operative conditions over the site of proposed treatment
    - Aortic Aneurysm

MYOFASCIAL RELEASE PROCEDURE

Diagnostic assessment of the specific somatic dysfunction (SD) or fascial pattern

Positioning into the direction of ease/barrier of the SD or traumatic pattern in as many planes of motion are possible

Apply an activation force (generally inherent force, physician-guided force, and/or respiratory cooperation)

Hold and monitor tissues until release is expressed. Characteristics of change in the tissues include: warmth, softening, increase in motion, palpation of fluid flow, or improved presence/vitality of primary respiratory mechanism

Reposition as necessary until normal tissue balance is achieved

Recheck

Video: Thoracic Inlet MFR

ARTICULATORY/SPRINGING

2 variations

Bringing a joint through ROM with low velocity and a moderate amplitude (LVMA). The motion may be done as a single motion or repeated several times until the range of motion is increased.

Moving joint into the restrictive barrier and then move back from the restrictive barrier and then re-engage the barrier. The action is repeated until there is an improvement in the ROM.

Video: Galbreath technique

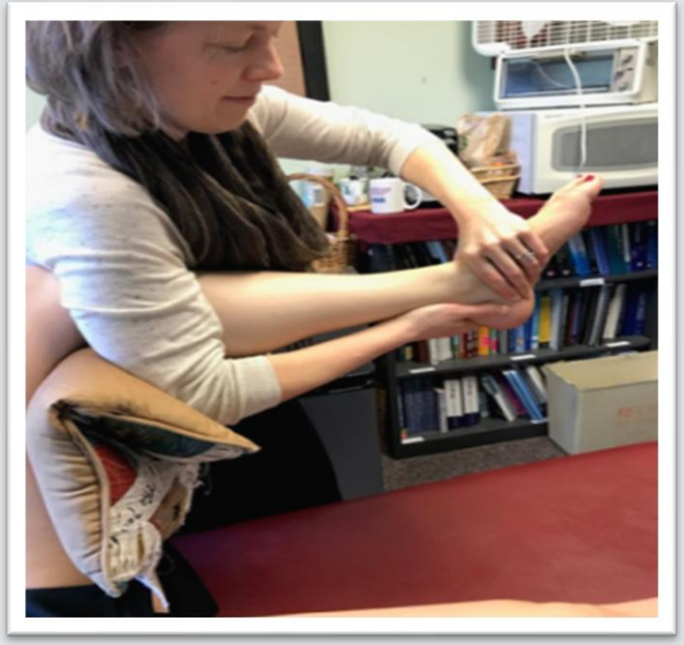
ARTICULATORY/SPRINGING

Indications:

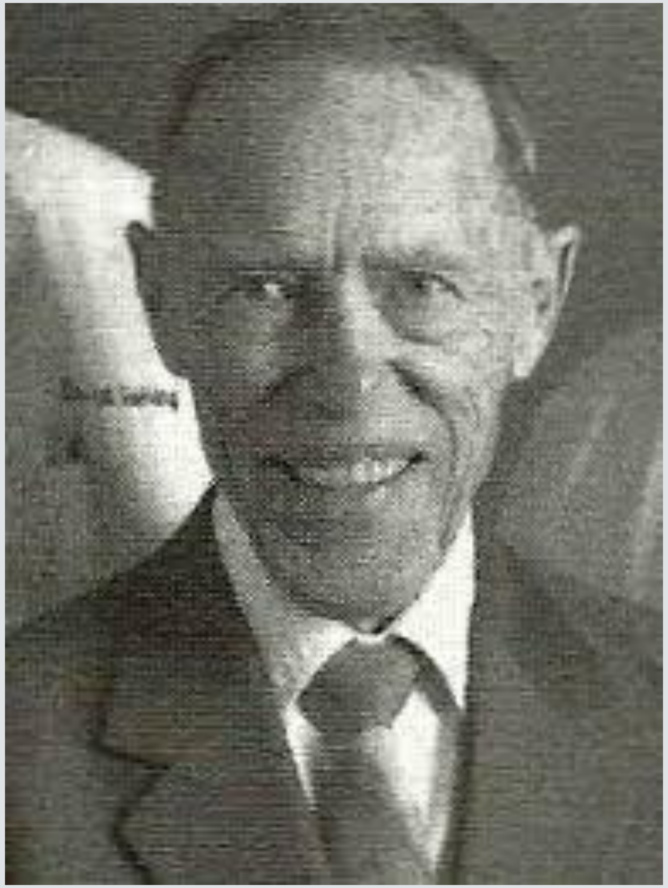
- Decreased articular range of motion
- Decreased gas exchange in lungs and fluid (lymph, blood) movement

Contraindications:

- Fracture in the area technique to be applied
- Septic joint



COUNTERSTRAIN



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Developed:

Lawrence Jones D.O.

Passive

Indirect

COUNTERSTRAIN

"A system of diagnosis and treatment that considers the dysfunction to be a continuing, inappropriate strain reflex, which is inhibited by applying a position of mild strain in the direction exactly opposite to that of the reflex; this is accomplished by specific directed positioning about the point of tenderness to achieve the desired therapeutic response."

"Counterstrain points are palpated as tense, somewhat edematous areas of tissue about the size of a fingertip. They are located in specific, predictable anatomical locations. Counterstrain points are exquisitely tender (about 4X normal) locally to an amount of pressure that normally would not elicit tenderness. The tissue changes of the Counterstrain points may range from several inches to virtually nonexistent. One or more Counterstrain points may be associated with a specific somatic dysfunction. "

NEUROMUSCULAR REFLEX

Protect Stability of Muscle.

Muscle Spindle:

Muscle Length.

Rate of change in muscle length.

Protect muscle from overstretching (muscle tearing).

Golgi Tendon:

Muscle Tension.

Rate of change in muscle tension.

Protects attachment portion of muscle (tearing off bone/avulsion).

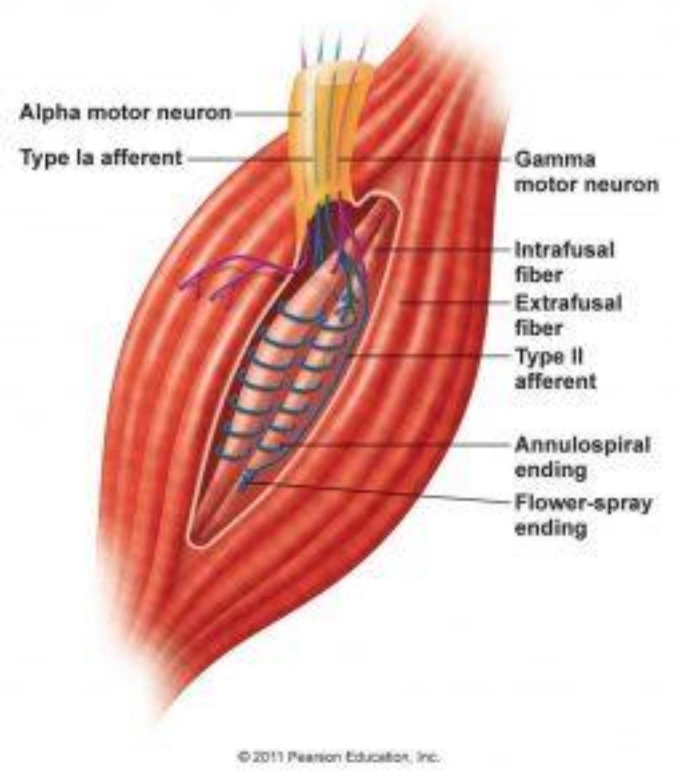
MUSCLE SPINDLE REFLEX

Intrafusal Mechanoreceptors within muscle belly of skeletal muscles.

Parallel to extrafusal skeletal muscle fibers.

Intrafusal Fibers :

- Nuclear Chain (2-5)
- Nuclear Bag (6-10)



https://www.velocity-pt.co.uk/wp-content/uploads/2016/11/muscle-spindle-254x300.jpg

MUSCLE SPINDLE REFLEX

Sensory Afferents:

Ia (primary endings):

Innervate every intrafusal fiber.

Surround the center of the spindle like a coil *(annulospiral endings).*

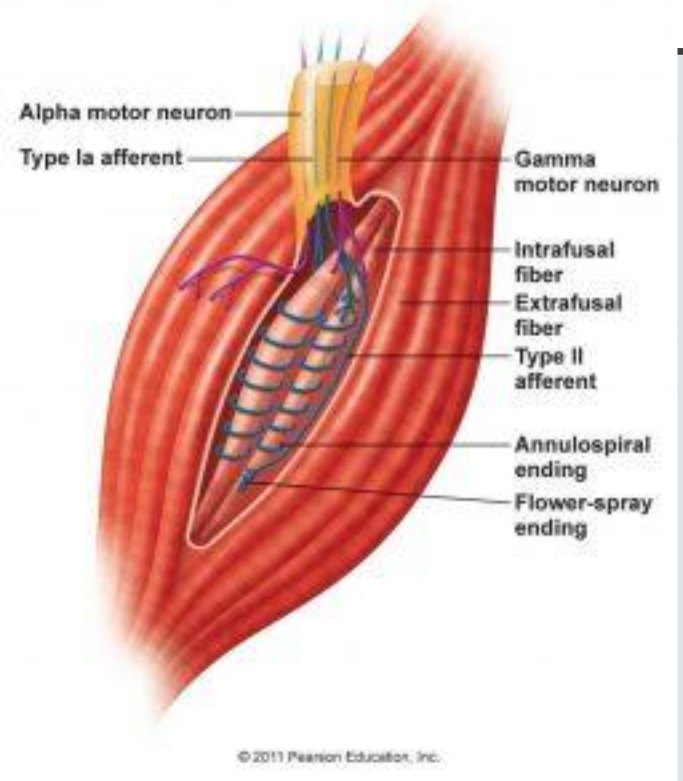
Primarily rate of change.

II (secondary endings):

Nuclear chain endings only.

Terminate on either side of Ia fibers in branch-like fashion *(flower-spray endings).*

Primarily length of change.



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MUSCLE SPINDLE REFLEX

Motor Efferents:

Maintenance of steady state vulnerable to overregulation.

Sudden stretch can result in powerful muscle contraction (inappropriate stimulation of gamma motor neurons).

Alpha motor neurons:

Innervate extrafusal fibers.

Results in contraction of shortening of whole muscle unit.

Regulated by both muscle spindle and golgi tendon

Gamma motor neurons (up to 70% 0f motor neurons):

Firing of gamma motor neurons brings Ia sensory fibers closer to their firing threshold

Gamma motor neurons mediate firing of fibers within the muscle spindle and regulate alpha motor neuron response

Normal gamma motor gain produces appropriate tone. Excessive gamma motor gain produces an inappropriate stretch reflex.

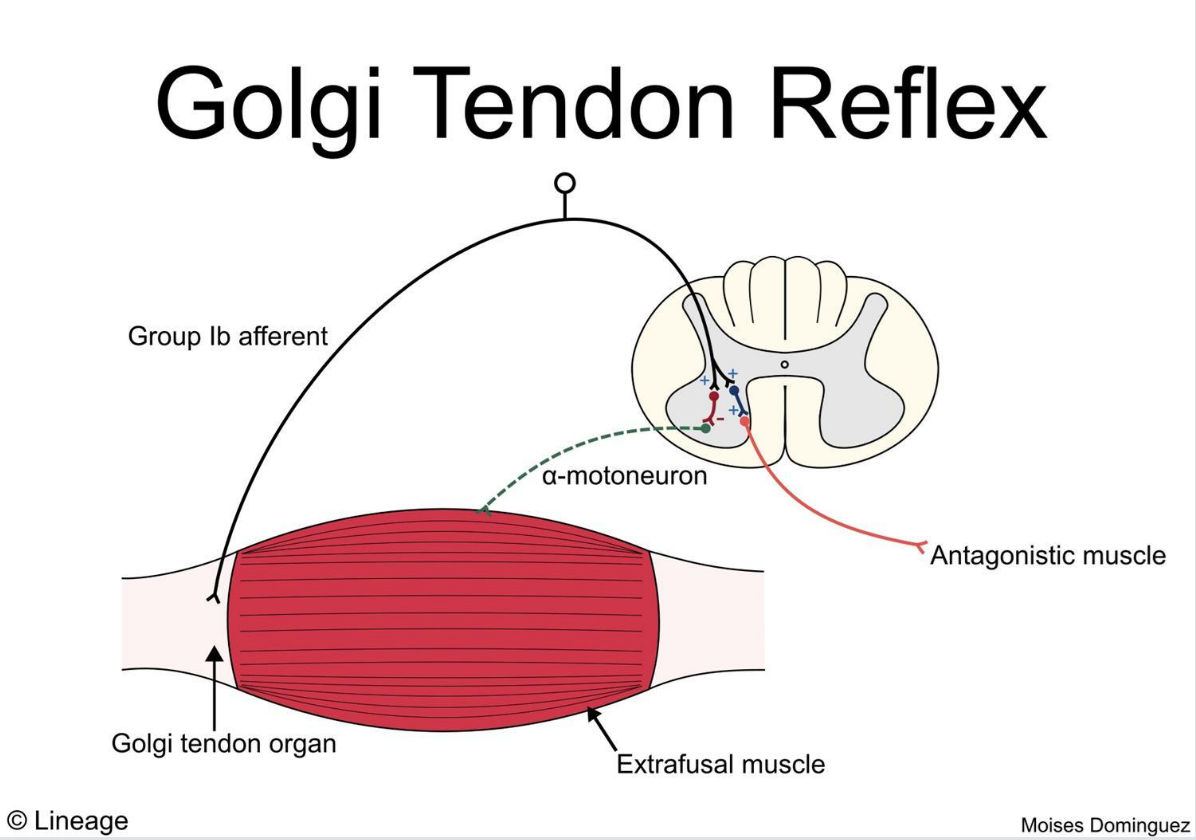
MUSCLE ENERGY PHYSIOLOGY- GOLGI TENDON REFLEX

Afferent Neurons:

Ib Group: Myelinated.

Synapse with inhibitory interneurons (and interneurons that ascend to higher CNS).

Inhibitory interneurons synapse with large alpha motor neurons.



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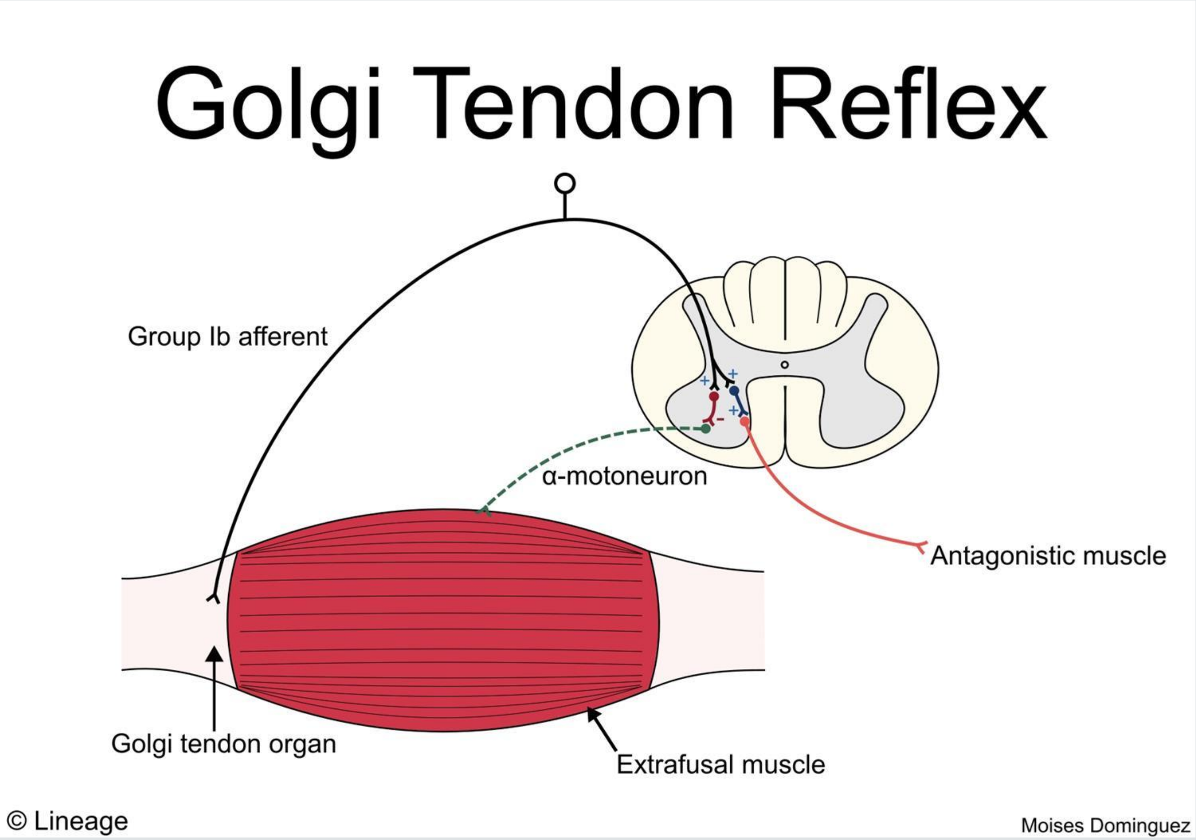
MUSCLE ENERGY PHYSIOLOGY- GOLGI TENDON REFLEX

Efferent Neurons:

Alpha motor neurons:

Inhibited by inhibitory interneurons.

Results in reflex relaxation of the muscle (same muscle that the Ib afferents originate).



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COUNTERSTRAIN

General Considerations:

Accurate DIAGNOSIS!

Tender points correlate with specific ligament, joint and muscle has it’s own tender point.

Indications:

Muscle hypertonicity

Somatic Dysfunction (Type I or II)- presumes SD have neuromuscular basis.

Contraindications:

Absolute: lack of patient consent, inability to tolerate classic treatment position (position may be modified), neurologic symptoms brought on by position, exacerbation of life-threatening symptoms due to positioning.
Relative: patient who cannot relax or is severely ill, extreme motions in the cervical spine in a patient with known vertebral artery disease or upper cervical ligamentous instability, malformation or osteoporosis, patient apprehension when approaching the treatment position

COUNTERSTRAIN

Procedure:

1. Identify the most significant Counterstrain point.
2. Establish a tenderness scale: 10-point scale. 0 = no tenderness, 10 = starting tenderness.
3. Monitor and retest Counterstrain point throughout the treatment.
4. Position the patient to eliminate tenderness.
5. Hold the position for 90 seconds (recheck every 30 seconds- maintain contact with point throught out)
6. Slowly return to the starting position without the help of the patient.
7. Recheck the Counterstrain point.

Goal: 2/3 reduction in tenderness.

Video: https://www.youtube.com/watch?v=fzGy96MqKIc

FACILITATED POSITIONAL RELEASE (FPR)

"A treatment method in which a dysfunctional body region is addressed with a combination of neutral positioning, application of an activating force (compression, torsion, or distraction), and placement into position of ease. A technique developed by Stanley Schiowitz, DO. (Glossary of Osteopathic Terminology, 2017)"

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FPR

- Indications:
- Somatic dysfunction of a facilitated segment
Autonomic dysregulation

- Contraindications:
Absolute: Lack of patient consent or cooperation.
Relative:
Infection, hematoma, or tear in involved muscle
Fracture or dislocation of involved joint
Rheumatologic conditions causing instability of the cervical spine
Undiagnosed joint swelling of the involved joint

FPR

1. Screen the region for area(s) of TART (remember to assess all four components, select specific areas of somatic dysfunction, obtain a segmental diagnosis of somatic dysfunction
2. Position physician and patient as appropriate for the region being treated and technique employed
3. Alter the patient’s natural position in order to produce a neutral spine in the region of the somatic dysfunction
4. *Move the segment into the position(s) of ease in all planes
5. **Induce a compressive force to the level of the dysfunction
6. Expect nearly immediate release of the articular restriction
7. Hold for about 3-5 seconds, return to original position
8. Re-assess somatic dysfunction and tissue quality

*It is equally valid to reverse steps 4 and 5
**The activating force may also be traction or torsion

Video: https://www.youtube.com/watch?v=r9VRbYTYkYs

STILL’S TECHNIQUE

A treatment technique that is “1. Characterized as a specific, non-repetitive articulatory method that is indirect, then direct. 2. Attributed to A.T. Still. 3. A term coined by Richard Van Buskirk, DO, PhD..”

(Educational Council on Osteopathic Principles (ECOP) Glossary of Osteopathic Terminology (GOT), 2017)

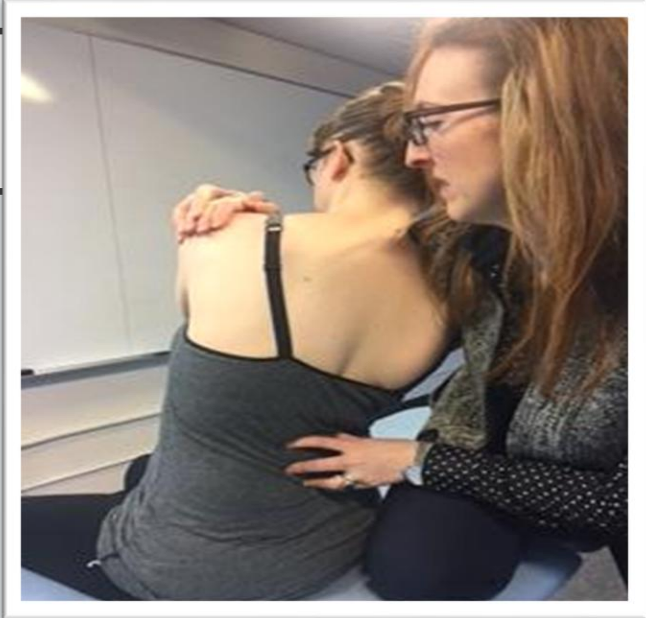
STILL TECHNIQUES

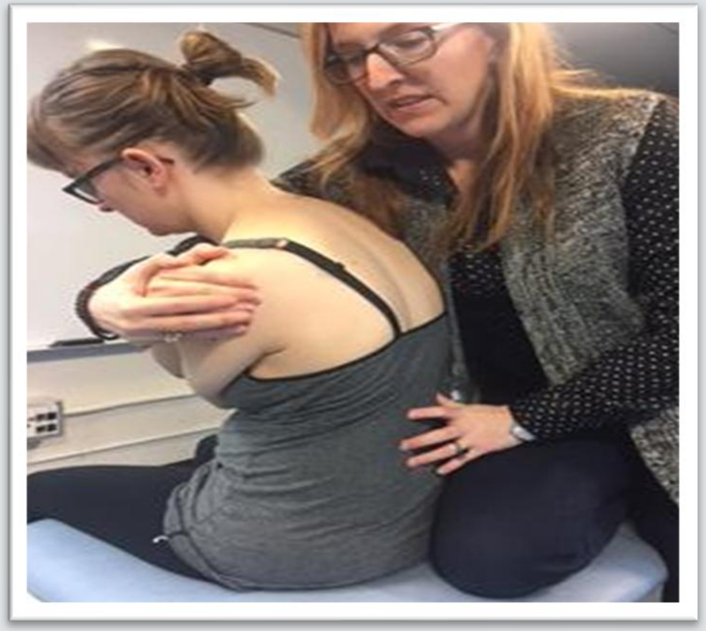
The Still Technique includes the following steps:

1. Evaluate the affected tissue and place it in its position of ease
2. Introduce a force vector to the affected tissue from another part of the body. This compression or traction should be less than 5 lb (2 kg)
3. Use the force vector to move the affected tissue initially into a smooth path to its position of ease and then alter the position while maintaining this force toward and through to the position of the restriction of position and movement
4. As the tissue moves through its restriction, a "bump" and/or a click may be felt or heard. Neither is necessary for correction of the somatic dysfunction
5. The region is passively moved back to neutral and retested

Video: https://www.youtube.com/watch?v=Ry6GCjWjG5Y

STILL TECHNIQUE





Indications:
Decreased range of motion
Somatic dysfunction

Contraindications:
Fracture at the site of somatic dysfunction
Septic joint
Fresh wounds at site of somatic dysfunction

BALANCED LIGAMENTOUS TENSION (BLT)

"According to Sutherland's model, all the joints in the body are balanced ligamentous articular mechanisms. The ligaments provide proprioceptive information that guides the muscle response for positioning the joint, and the ligaments themselves guide the motion of the articular components." (Foundations & Glossary of Osteopathic Terminology)

BLT

For any given joint, the type and amount of normal physiologic motion is dependent on the shape of the articulatory surfaces, the muscles which act on that joint, and the position of the ligaments stabilizing it. Because the ligaments of a normally functioning joint exist in a balanced, reciprocal tension, a disruption in that balance can cause and/or maintain somatic dysfunction. When a ligament is strained (by injury, inflammation, mechanical forces, etc.), the distribution of tension affecting the joint becomes altered. Some of the ligaments will be stressed while others become lax.

To treat somatic dysfunction using Balanced Ligamentous Tension (BLT), the joint must be positioned so that the forces in the ligaments of the joint once again become balanced. The joint is moved in the direction of the strained ligament(s) until tension within those ligaments becomes equal to those which are not strained. At this point, balanced tension is achieved. Using compression/distraction and/or respiratory assistance can also facilitate treatment. Hold the position for several respiratory cycles until you feel a release.

BLT

Principles of Balanced Ligamentous Tension:

1. Compression/distraction forces are referred to in Sutherland's writings but are not necessarily performed as a distinct step in the process of balancing ligamentous tensions for all techniques.

2. Breath may be used as an additional activation force.

BLT

Indications:

Somatic dysfunction

Contraindications:

Absolute: Lack of patient consent or cooperation

Relative:
Infection, hematoma, or tear in involved muscle
Fracture or dislocation of involved joint
Positioning that compromises vasculature
Open wound in area of dysfunction

OCMM

A system of diagnosis and treatment by an osteopathic physician using the primary respiratory mechanism and balanced membranous tension.

Based on the five components of the primary respiratory mechanism:

1. Inherent motility of the brain and spinal cord
2. The rhythmic fluctuation of the CSF
3. The motion of the intracranial and intraspinal (dural) membranes
4. The articular mobility of the bones of the cranium
5. The articular mobility of the sacrum between the ilia

OCMM

Principles of treatment:

Balanced membranous tension - achieving a balanced tension within dural and/or other connective that can be executed utilizing direct or indirect direction of force
Exaggeration - often used when addressing sphenobasilar synchondrosis strain patterns
Directing of fluid (sometimes referred to as the tide) - directing fluctuant fluid forces through gentle pressures on the cranium and/or associated structures
Direct or molding - applied direct forces typically to cranial vault structures
Disengagement (articular/suture release) - specific decompressive forces applied to sutural restrictions, often used in tandem with fluid as noted above

Video (1:47): htthttps://www.youtube.com/watch?v=rDr9RlgkhUc

OCMM

Indications:

OCMM is used to treat somatic dysfunction in the head region. Clinical presentations that may have relevant somatic dysfunction include:

Bell's palsy
Cranial neuropathy- nerve entrapment
Colic
Headache
Orofacial pain
Otitis Media

Sinusitis
TMJ
Vertigo
Feeding difficulties
Plagiocephaly
Torticollis
Trigeminal neuralgia
Tinnitus

OCMM

Contraindications:

Absolute:

- Intracranial bleeding
- skull fracture
- acute cerebrovascular accident

Relative:

- Coagulopathies
- Space occupying lesion in cranium
- Increased intracranial pressure

LYMPHATICS TECHNIQUES

Techniques used to optimize function of the lymphatic system – Glossary of Osteopathic Terminology, 2017.

Lymphatic techniques do not encompass a specific modality; rather they include techniques which are intended to encourage lymphatic flow and/or remove impediments to lymphatic circulation. Techniques that augment lymphatic flow typically rely on mechanical movement (thoracic pump, pedal pump, etc.). In contrast, treatments intended to remove obstruction to lymphatic flow typically address the transverse diaphragms and fascial restrictions of the body. When treating lymphatics, it is imperative that you first open the terminal drainage of the lymphatic system by treating the thoracic inlet.

LYMPHATIC TECHNIQUES

Key Concepts:

The lymphatic system plays a major role in numerous homeostatic mechanisms of the body.
The lymphatic system is particularly vulnerable to fascial dysfunction in the region of the superior thoracic inlet.
Initial draining of the lymphatic system must always be done through terminal areas.
The risk-to-benefit ratio for lymphatic techniques is quite good. No known significant complications have resulted from osteopathic lymphatic treatment to date. Clinical judgment, however, should always be used when prescribing treatment for a patient in any disease state.

LYMPHATIC TECHNIQUES

Indications:

Acute somatic dysfunction
Sprains, strains, etc.
Edema
Tissue congestion
Lymphatic/venous stasis
Pregnancy
Infection
Inflammation
Pathologies with significant venous and/or lymphatic congestion

Contraindications:
Absolute:
Anuria (if patient isn't on dialysis)
Necrotizing Fasciitis
Local Fracture
Lack of patient consent

Relative:
Inability of CHF patient to tolerate excessive preload
COPD (thoracic pump with activation only) due to increased residual volume post treatmenti

- Foundations of Osteopathic Medicine, 3rd edition

Video (2:50): https://www.youtube.com/watch?v=YB4pjIJflU4&feature=emb_logo

VISCERAL TECHNIQUES

All of the viscera move with each breath (mobility) → necessary to keep fluid and pressure distribution within normal limits

This mobility can be altered just like in the MSK system

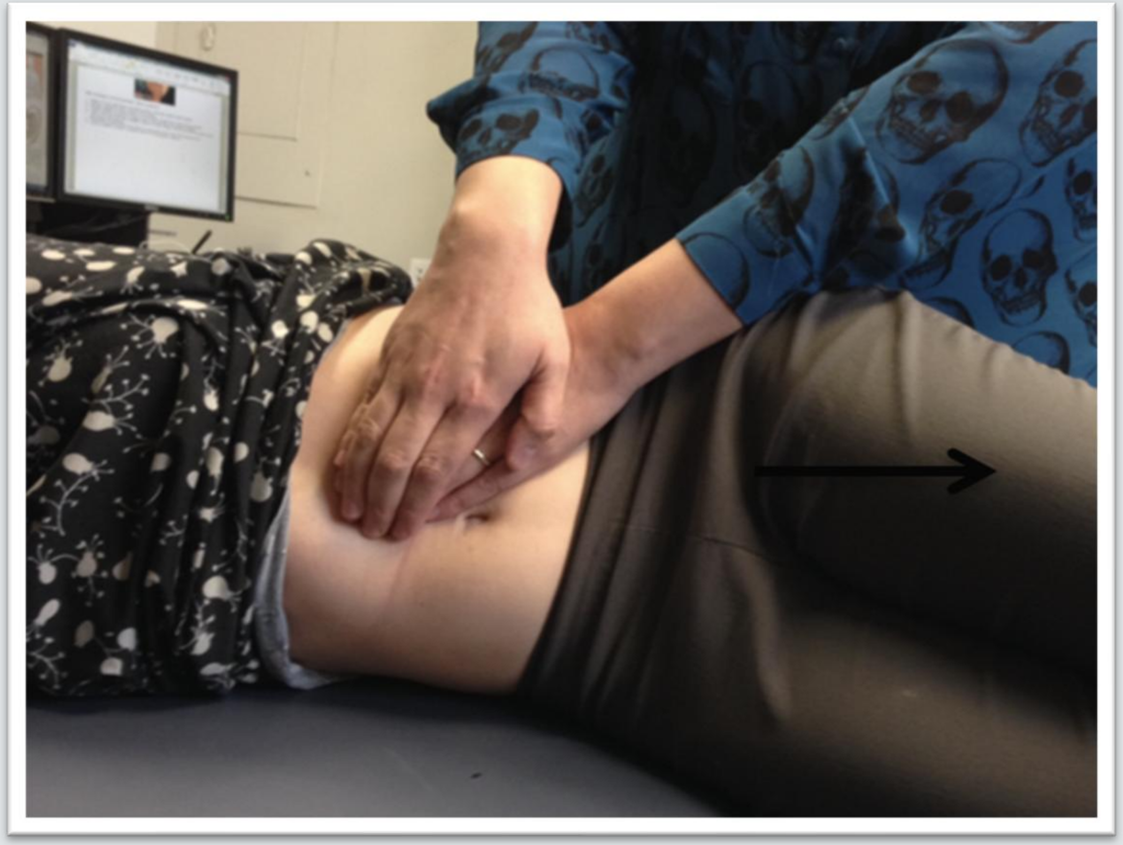
Altered mobility → pathology/symptoms

Examples of what you might treat: asthma (lungs), recurrent UTIs (kidneys), chronic alcohol use (liver), BPH, low back pain, flank pain, heartburn (GERD), constipation, etc

VISCERAL OMT APPROACHES

- Indirect/exaggeration
- Direct
- With respiratory assistance
- Lymphatic
- Vascular
- Counterstain
- Spinal/cranial

Video (1:04): https://www.youtube.com/watch?v=zV-rnEl4D5A



VISCERAL CONTRAINDICATIONS

- High fever of unknown origin
- Traumatic internal bleeding
- Non medically treated cancer
- Infectious diseases not medically controlled
- Unstable medical problems without a proper diagnosis

MORE HIGH-YIELD CONTENT

Organ	Sympathetic Innervation	Parasympathetic Innervation
Head and Neck	T1- T5	Vagus (also CNIII, VII, IX)
Heart	T1- T5	Vagus
Lung	T2-T6	Vagus
Lower Esophagus / Stomach	T5-T10	Vagus
Liver and Gallbladder	T6-T9	Vagus
Small Intestine	T10-T11	Vagus
Ascending / Transverse Colon	T12-L1	Vagus
Descending and Sigmoid Colon / Rectum	L1-L2	S2-S4
Kidney	T10-L1	Vagus
Ureter	T10-L2	Vagus (proximal 2/3), S2-S4 (distal 1/3)
Ovary/Testes	T9-T11	S2-S4
Bladder	T10-L2	S2-S4
Uterus	T12-L1	S2-S4
Cervix	T10-L2	S2-S4

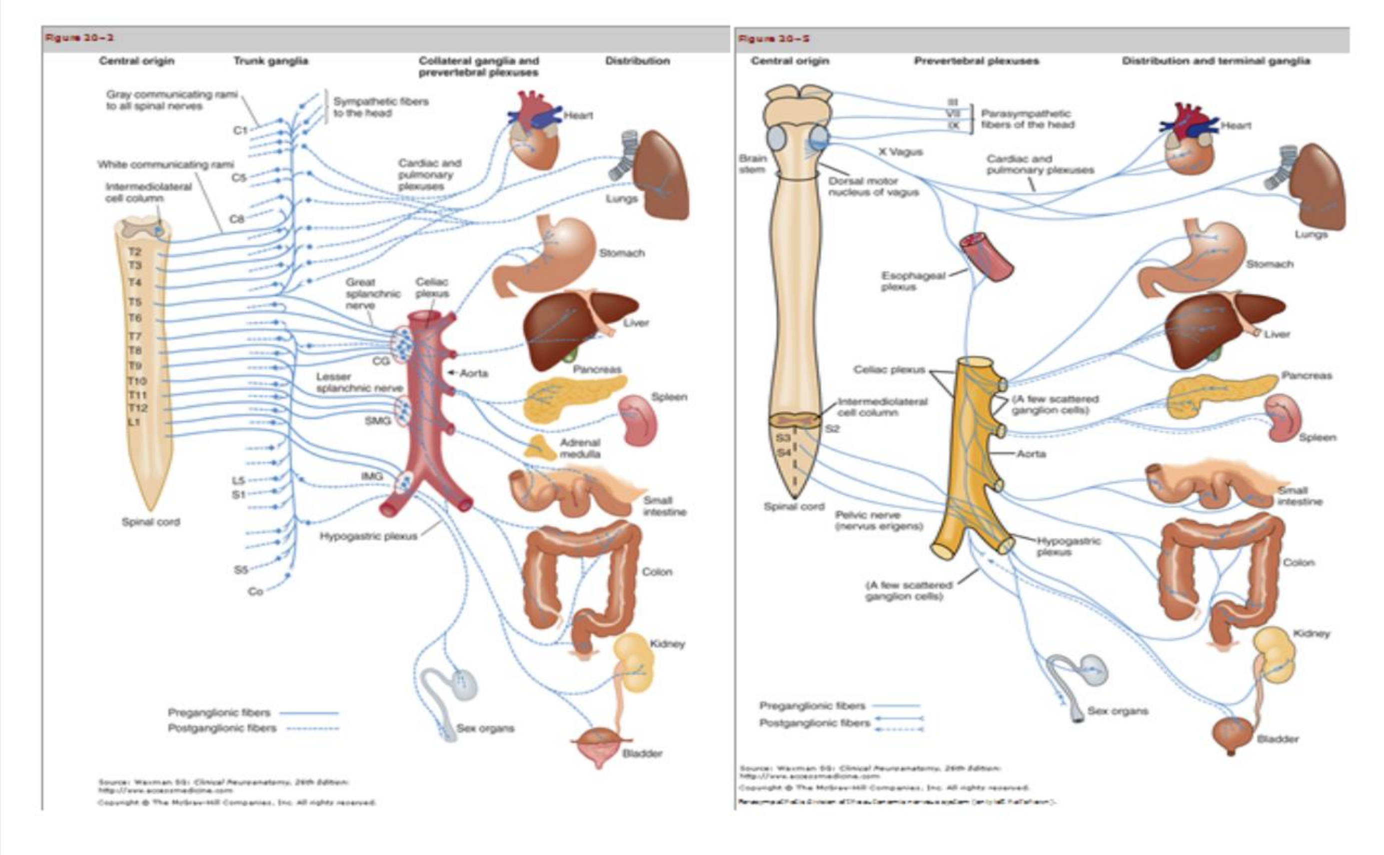
VISCEROSOMATIC REFLEXES *FROM TUCOM ANS TBL IN OPP I

Viscerosomatic Reflexes

Visceral Organ	Spinal Cord Level	Corresponding Nerve & Ganglion
Head and Neck	T1-T4	
Heart	T1-T5	
Respiratory System	T2-T7	
Esophagus	T2-T8	
Upper GI Tract Stomach Liver Gallbladder Spleen Portions of pancreas & duodenum	T5-T9	Greater Splanchnic Nerve Celiac Ganglion
Middle GI Tract Portions of pancreas & duodenum Jejunum Ilium Ascending colon & proximal 2/3 of transverse colon	T10-T11	Less Splanchnic Nerve Superior Mesenteric Ganglion
Lower GI Tract Distal 1/3 of transverse colon Descending colon & sigmoid colon Rectum	T12-L2	Least Splanchnic Nerve Inferior Mesenteric Ganglion

Visceral Organ	Spinal Cord Level	Corresponding Nerve & Ganglion
Appendix	T12	
Kidneys	T10-T11	Superior Mesenteric Ganglion
Adrenal Medulla	T10	
Upper Ureters	T10-T11	Superior Mesenteric Ganglion
Lower Ureters	T12-L1	Inferior Mesenteric Ganglion
Bladder	T11-L2	
Gonads	T10-T11	
Uterus & Cervix	T10-L2	
Erectile tissue of penis & clitoris	T11-L2	
Prostate	T12-L2	
Arms Legs	T2-T7 T10-L3	

Savarese OMT Review

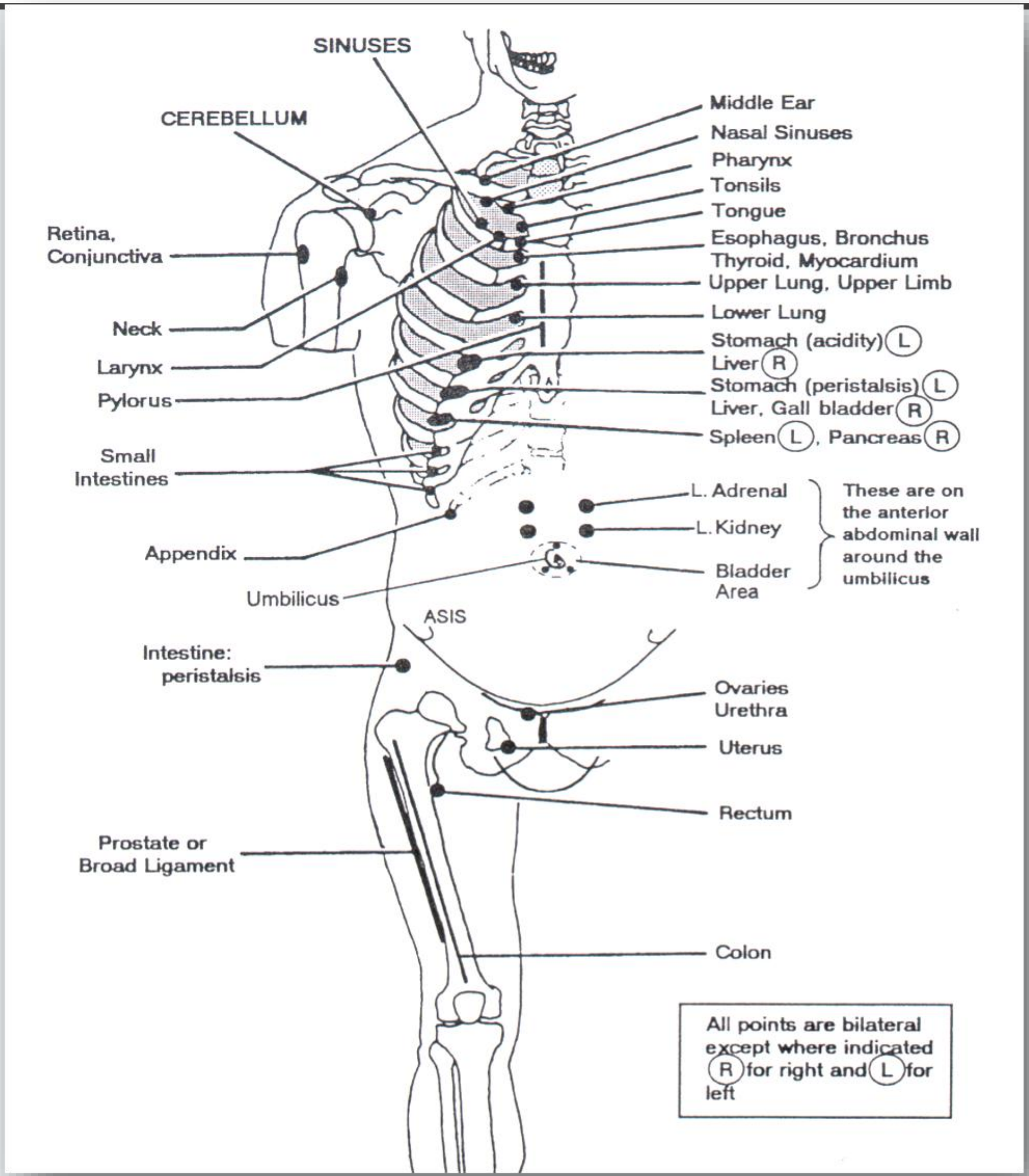


VISCEROSOMATIC REFLEXES

CHAPMAN'S POINTS

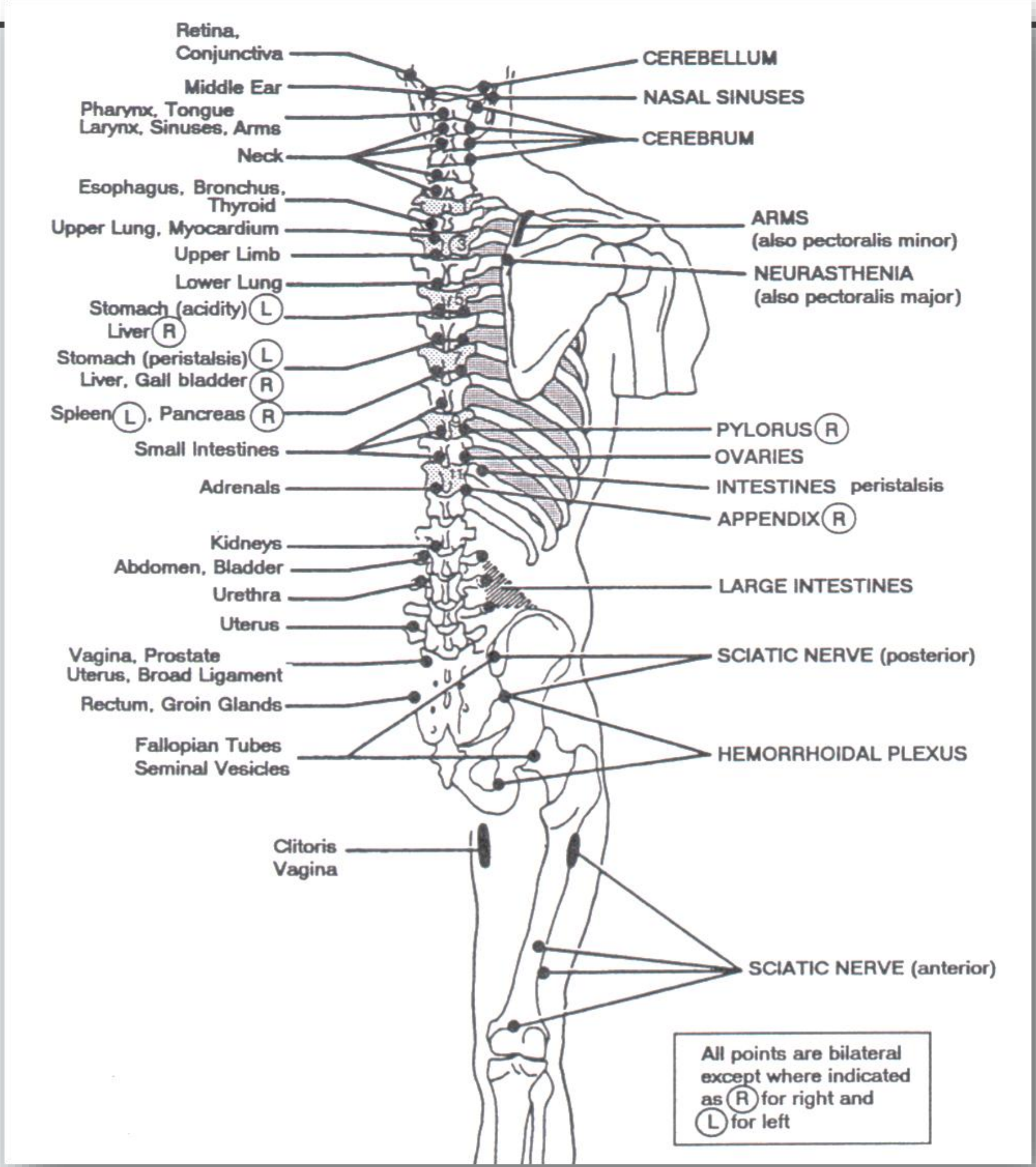
- Neurolymphatic reflex
- Anterior and posterior points to screen and treat

ANTERIOR POINTS



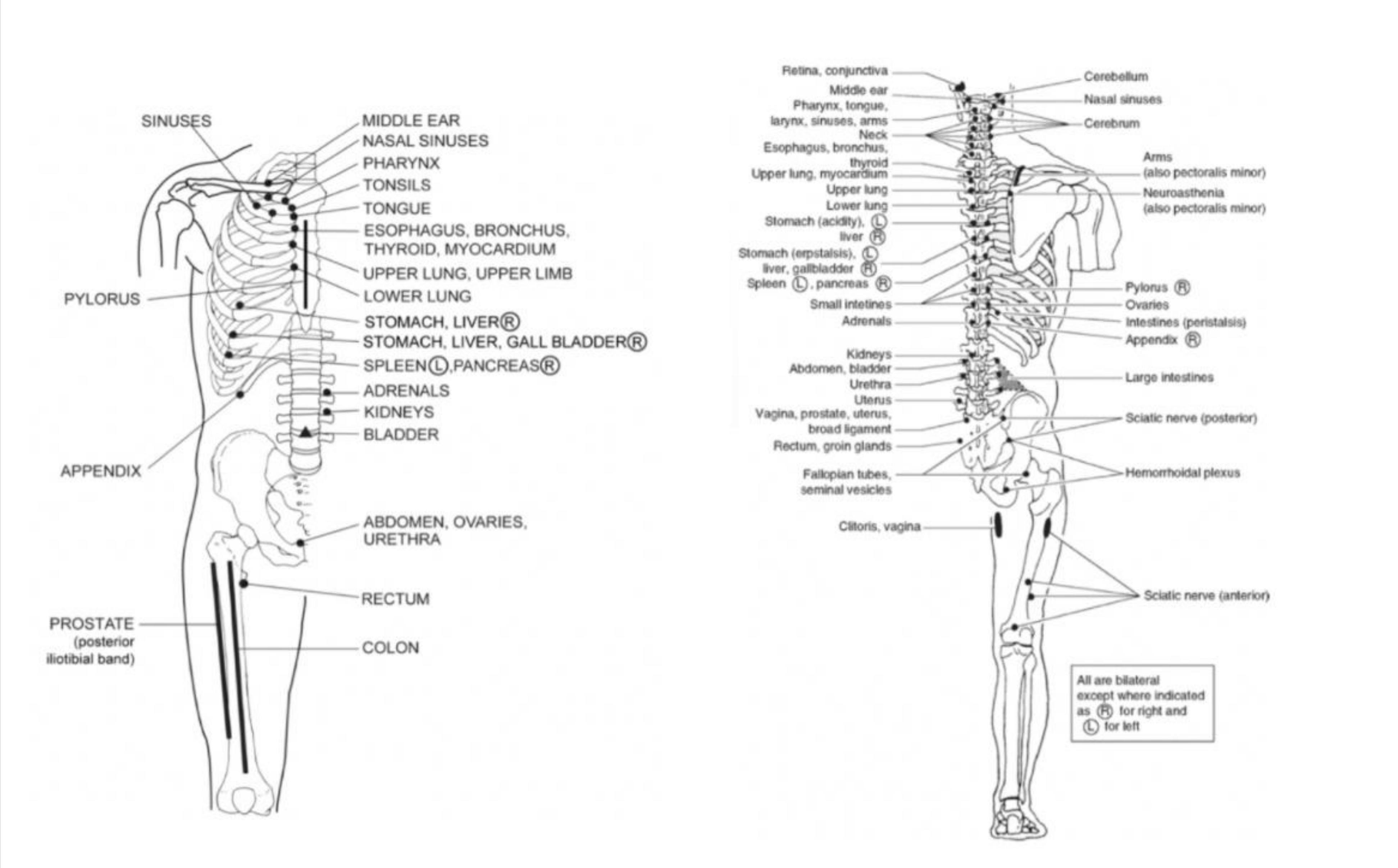
Osteopathic Considerations in Systemic Disease, 2nd Ed pp 232-233

POSTERIOR POINTS

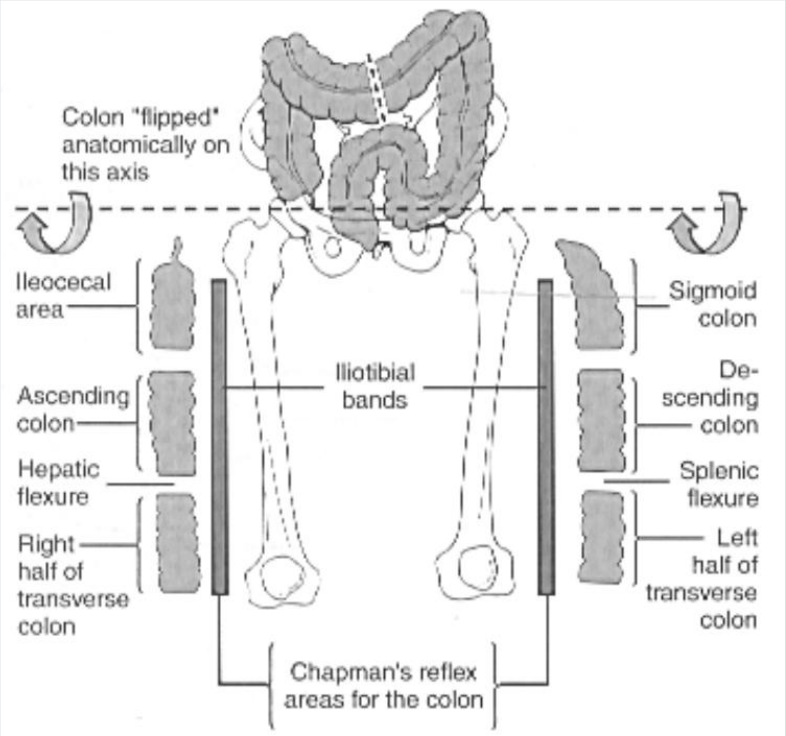


Osteopathic Considerations in Systemic Disease, 2nd Ed pp 232-233

Chapman's Points



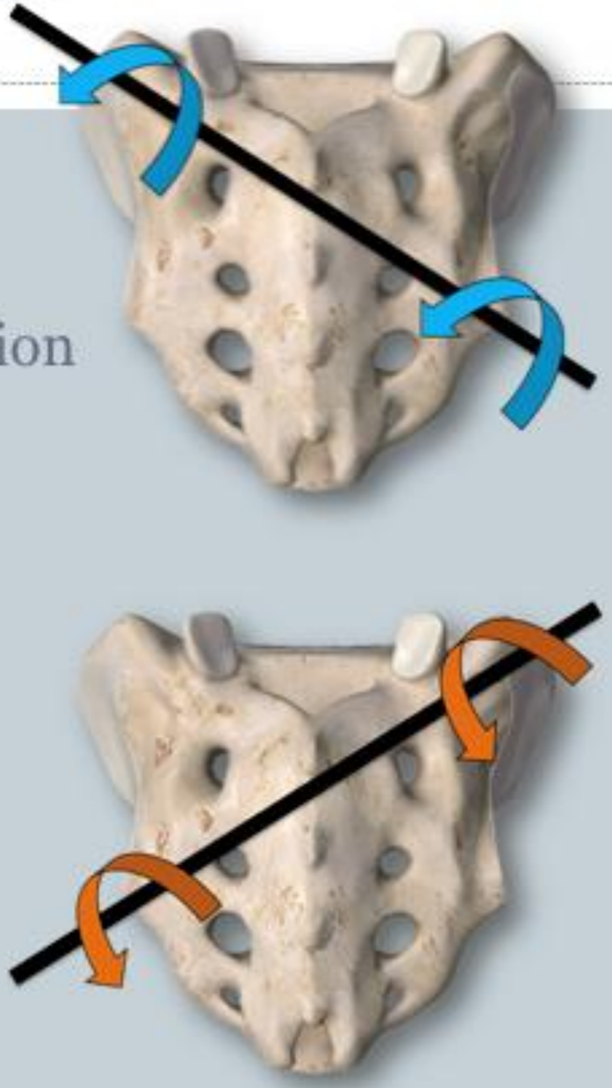
Location	Anterior	Posterior
Upper respiratory tract	Near clavicle, rib 1 or 2	C1 or C2
Lower respiratory tract Bronchi Upper lung Lower lung	 Rib 2 & 3 at sternocostal junction Rib 3 & 4 at sternocostal junction Rib 4 & 5 at sternocostal junction	 T2 T3/4 T4/5
Upper GI tract Esophagus Stomach Liver, gallbladder Pancreas Spleen Duodenum Jejunum Ilium	 Rib 2 & 3 at sternocostal junction Ribs 5-7 left Ribs 5-7 right Ribs 7 & 8 at costochondral junction right Ribs 7 & 8 at costochondral junction left Ribs 8 & 9 near costochondral junction Ribs 9 & 10 near costochondral junction Ribs 10 & 11 near costochondral junction	 T2 T5-7 (left) T5-7 (right) T7-8 (right) T7-8 (left) T8,9 T9,10 T10,11
Lower GI tract Appendix Colon	 Tip of rib 12 right See figure below	 TP's of T11/T12 L2-L4 to iliac crest
Heart	Between ribs 2&3 at sternocostal junction	T2/3
Urinary system Kidney Bladder Urethra	 1" above umbilicus either side of midline Periumbilical and pubic symphysis Pubic symphysis	 T12, L1 TP of L2 TP of L2
Reproductive organs Ovaries/testes Uterus Prostate	 Anterior pubic bone Medial border of obturator foramen Lateral femur (similar to colon)	 T9-T11 TP of L5 TP of L5 to PSIS



In general anterior points are located at the sternocostal or costocondral junction unless otherwise specified in the table above, and posterior points are located between the spinous process and transverse process of the vertebrae. For example, if the table above states the posterior point is T2, that implies the Chapman's point is midway between the spinous process and transverse process of T2. Likewise T3/T4 implies between the transverse process of T3 and T4, midway between the transverse and spinous process.

Savarese OMT Review, http://files.academyofosteopathy.org/convo/2016/Handouts/Devine_SAAOLecture.pdf

Sacral Diagnosis



- Seated Flexions Test:
  - Opposite axis (oblique axis)
  - Same side as a unilateral flexion/extension (shear).
- Spring or Sphinx Test:
  - Positive: Backward (non-neutral)
  - Negative: Forward (neutral)
- Deep Sulcus
- Posterior/inferior ILA
- L5 Diagnosis
  - (??Sidebending sets the access)

Perform standing and seated flexion test

(+) Standing flexion test

IS Somatic Dysfxn

Examine 4 things:
- ASIS level
- PSIS level
- pubic symphisis level
- distance from ASIS to umbilicus

Which side is (+) (right or left)?

(+) on Rt.

(+) on Rt.
Rt. ASIS superior
Rt. PSIS superior
Rt. Pub Symph sup
Dx: Rt. Sup Shear
(whole innominate shifted superiorly)

(+) on Rt.
Rt. ASIS inferior
Rt. PSIS inferior
Rt. PS inferior
Dx: Rt. Inf Shear
(whole innominate shifted inferiorly)

(+) on Rt.
Rt. ASIS inferior
Rt. PSIS superior
Rt. PS inf (or sym)
Dx: Rt. Anterior Innom. Rotation

(+) on Rt.
Rt. ASIS superior
Rt. PSIS inferior
Rt. PS sup (or sym)
Dx: Rt. Posterior Innom. Rotation

(+) on Rt.
Rt. ASIS further from umbilicus (than lft ASIS)
Rt. PSIS more medial
PS symmetrical
Dx: Outflare on Rt.

(+) on Rt.
Rt. ASIS closer to umbilicus (than lft ASIS)
Rt. PSIS more lateral
PS symmetrical
Dx: Inflare on Rt.

(+) on Left.

(+) on Lft.
Lft. ASIS sup
Lft. PSIS sup
Lft. PS sup
Dx: Lft. Sup Shear
(whole innomimate shifted superiorly)

(+) on Lft.
Lft. ASIS inf
Lft. PSIS inf
Lft. PS inf
Dx: Lft. Inf Shear
(whole innominate shifted inferiorly)

(+) on Lft.
Lft. ASIS inf
Lft. PSIS sup
Lft. PS inf (or sym)
Dx: Lft. Anterior Innom. Rotation

(+) on Lft.
Lft. ASIS sup
Lft. PSIS inf
Lft. PS sup (or sym)
Dx: Lft. Posterior Innom. Rotation

(+) on Lft.
Lft. ASIS further from umbilicus (than lft ASIS)
Lft. PSIS more medial
PS symmetrical
Dx: Outflare on Lft.

(+) on Lft.
Rt. ASIS closer to umbilicus (than lft ASIS)
Rt. PSIS more lateral
PS symmetrical
Dx: Inflare on Rt.

(+) Seated flexion test

SI Somatic Dysfxn

Examine 5 things:
- Sacral sulcus depth
- ILA
- Spring test
- Backward bending test
- Position of L5 relative to S1

See next page

SI Somatic Dysfxn

(positive seated flexion test)

Examine 5 things:
- Sacral sulcus depth
- ILA
- Spring test
- Backward bending test
- Position of L5 relative to S1

If the **deeper sulcus depth** is on the **opposite** side as that of the more **posterior ILA**, then dysfxn is a **<u>torsion</u>** (ie. sacral sulcus deep on rt. & posterior ILA on lft.)

If the **deeper sulcus depth** is on the **same** side as that of the more **posterior ILA**, dysfxn is a **<u>sacral flexion or extension</u>** (ie. sacral sulcus deep on rt. & posterior ILA on rt.)

Things to remember for diagnosis of SI dysfunction (torsion or flexion/extension of sacrum):
- A **deep** sacral sulcus implies that the sacral base on that side is more **anterior** (flexed) than on the opposite side.
- Diagnosis is always in relation to the side with the positive seated flexion test.

Things to remember for <u>sacral torsion</u> diagnosis:
- The **oblique axis** (the axis that the sacrum rotates around) always **start on the opposite side of the (+) finding** (ie. if **positive seated flexion test is on the right**, then the axis is drawn from the lft. sacral base through the Rt. ILA & is considered a **left axis**)

- **For torsions**, a deep sacral sulcus means that the sacral base is going forward on that side, therefore if the **(+) seated flexion test** is found to be on the **same side as that of the deep sacral sulcus**, then **Dx is a forward sacral torsion**. (ex. (+) on Rt, deep sacral sulcus on right, posterior ILA on left … Dx: forward sacral torsion.) On the other hand, if the **sacral sulcus is deeper on the opposite side as the (+) finding**, that implies that the sulcus depth is more shallow on the side of the (+) finding, hence the sacral base is more posterior and is considered to be a **backward sacral torsion**.

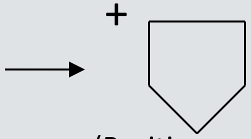
- L5 findings can either be a Type I or Type II mechanics (see section below) with **L5** being **rotated towards the side of the deeper sulcus depth**. (Ex. if sacral sulcus depth is deeper on the rt, then L5 will be rotated Rt, regardless of backwards or fwd torsion)

- For **backward sacral torsions**: (since it's backward, everything's bad… spring test and symmetry of sacral sulcus depth)
  - Spring test is positive (positive means that the sacrum does not exhibit springiness… it's like pressing on a rock)
  - Sacral sulcus depth gets less symmetrical in the backward bending test. Similarly, ILAs also get less symmetrical

- For **forward sacral torsions**:
  - Spring test is negative (good spring)
  - Sacral sulcus depth gets more symmetrical with the backward bending test. Similarly, ILAs also get less symmetrical

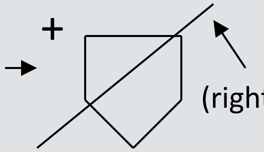
Steps for diagnosing SI Somatic Dysfxn:

1) Find out which side has (+) seated flexion test finding? (draw a sacrum with a (+) on appropriate side)



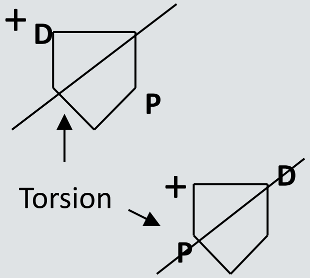
(Positive on left)

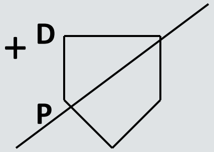
2) Determine what the axis is. (Axis always start on the opposite side of (+) finding)



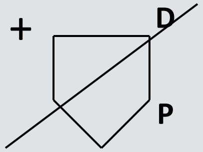
3) Determine which sacral sulcus is deeper & which ILA is more prominent (more posterior). Draw on sacrum. (D=deep, P=posterior)

4) Determine if the dysfunction is a torsion or a flexion/extension problem. (For **torsions**, **D (deep sacral sulcus) and P (prominent ILA)** are on **opposite sides**. For **flexion/extension**, **D and P are on the same side** and diagnosis is explained below)

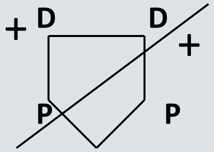




Dx: Left unilateral flexion (b/c on the (+) side, sulcus is deep, therefore positioned more forward & "flexed"



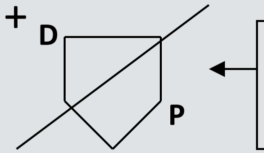
Dx: Left unilateral extension (b/c on the (+) side (left), sulcus is shallow, therefore positioned more posterior & "extended"



Dx: Bilateral flexion

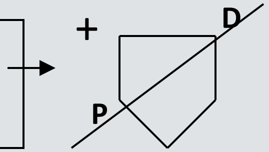
Note: If bilaterally positive seated flexion test, both sacral sulci are shallow and both ILAs are anterior, then Dx is Bilateral extension

5) If torsion, look at the sacral sulcus depth on the side with the (+) and determine if it's going forward (deep) or backwards (shallow) in comparison to the other side. Determine if it's a forward sacral torsion or backwards sacral torsion & diagnose based on the side of the (+) dysfunction.

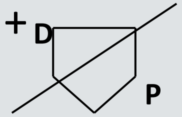


On the **(+) side**, sulcus depth is **deep**, therefore, going forward and is considered a **forward sacral torsion**.

On the **(+) side**, sulcus depth is **shallow**, therefore, more posterior (going backwards) & is considered a **backward sacral torsion**.

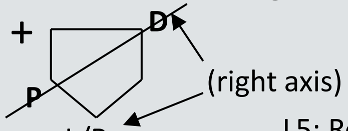


6) If forward sacral torsion is diagnosed, then the first and second letters of the diagnosis are the same, the second letter being the axis and the first letter being the side that the sacrum is facing (turning towards). (ex. if (+) finding is on the left and the left sacral sulcus is deeper, then axis is on the right, and the diagnosis is a R/R (right on right) forward sacral torsion. If a backward sacral torsion is diagnosed, then the two letters are not the same (with the axis being the second letter).



R/R

L5: Rotated left



L/R

L5: Rotated right

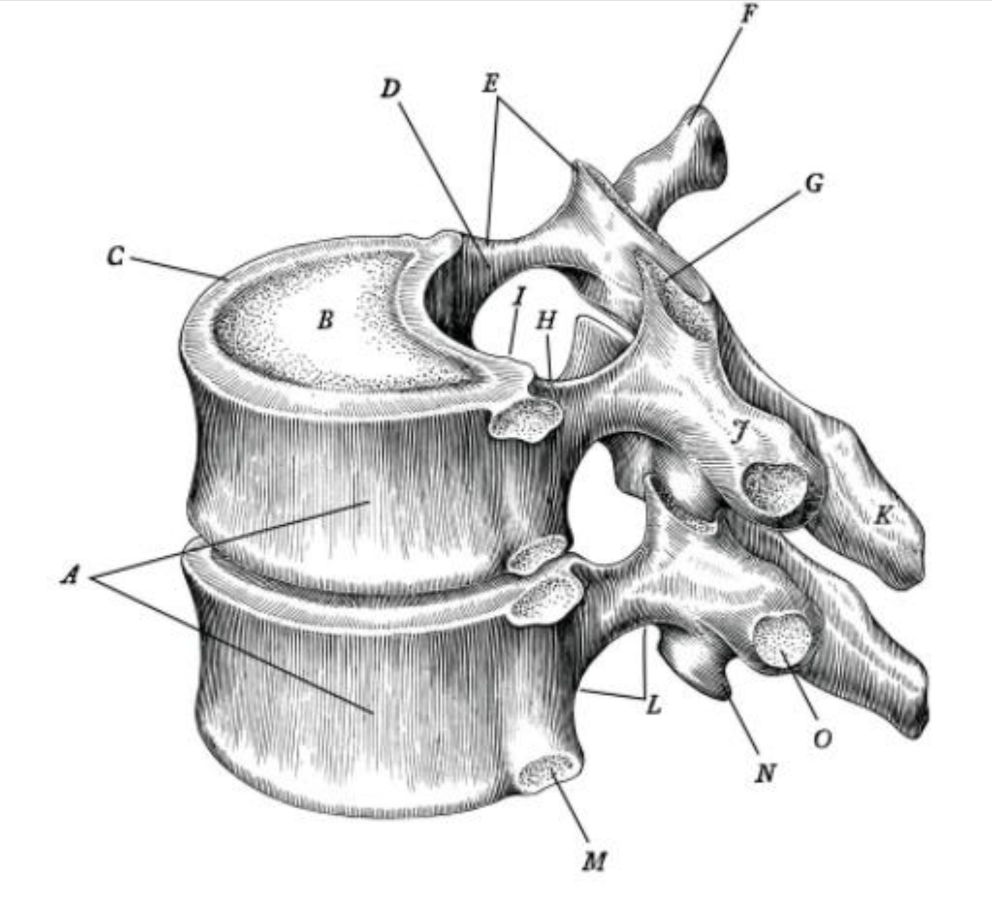
ME Tx for R/L Backward Sacral Torsion

ME Tx for L Unilateral Sacral Flexion

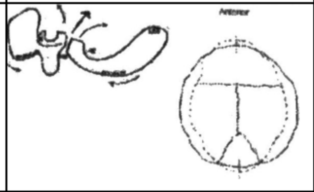
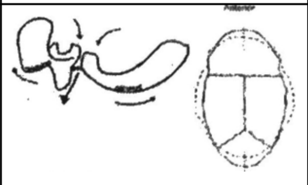
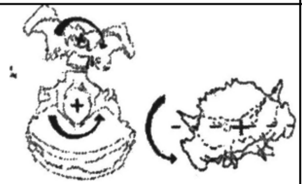
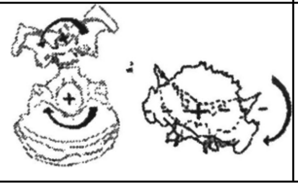
ME Tx for L/L Forward Sacral Torsion

Somatic dysfunction is named for the freedom of motion.

Vertebral segments are described <u>in relation to the vertebra below.</u>



https://www.vectorstock.com/royalty-free-vector/thoracic-vertebrae-anatomy-vintage-clip-art-vector-23244409

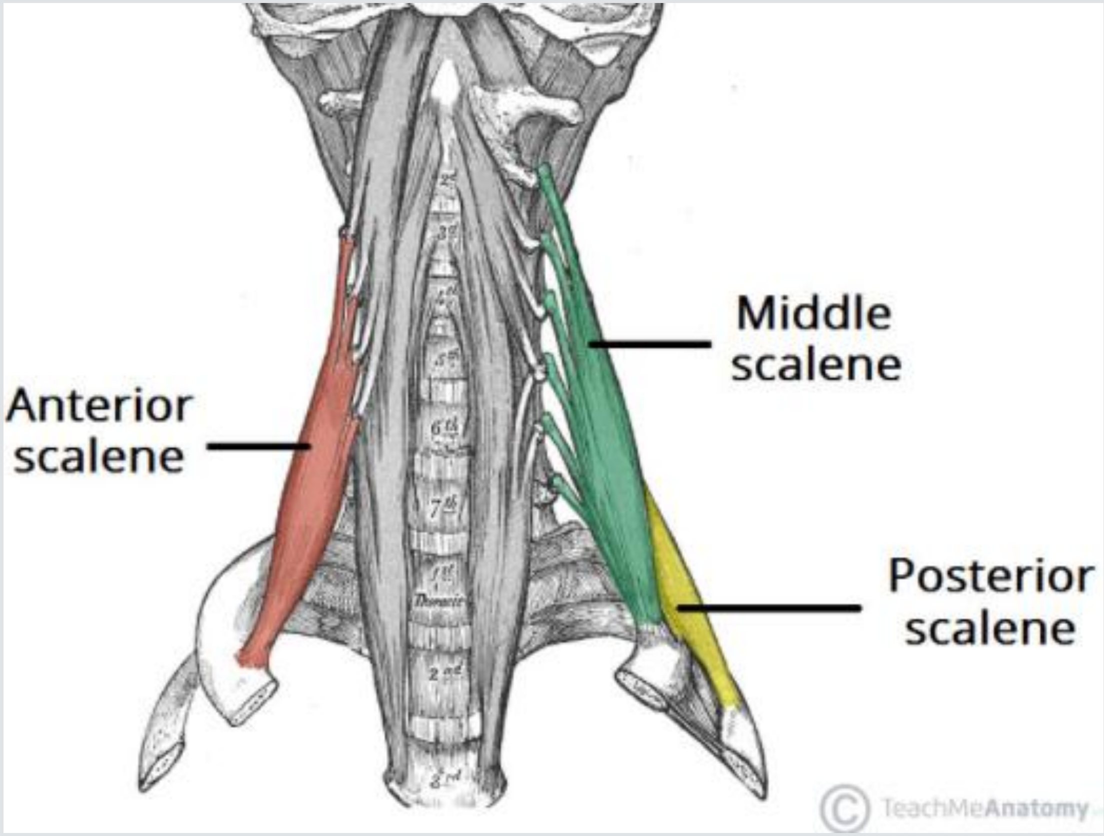
Strain Pattern	Axis	Mechanics	Motion of Bones	Vault Hold
Flexion	2 transverse axes (Rotation in opposite direction)	SBS moves superiorly		Narrowing of A-p diameter, widening of transverse diameter
Extension		SBS moves inferiorly		Widening of A-p diameter, narrowing of transverse diameter
Torsion	1 A-P axis (rotation in opposite direction)	<u>Left torsion</u> - left sphenoid greater wing moves superiorly		Left torsion - left index finger moves superiorly
		<u>Right Torsion</u> right sphenoid greater wing moves superiorly		Right torsion right index finger moves superiorly
Sidebending rotation	2 vertical axes (rotation in opposite directions) and 1 A-P axis (rotation in same direction)	<u>Left Sidebending Rotation</u> - left sided convexity		Left sided convex left side gets wider and fuller
		Right Sidebending <u>Rotation</u> - right sided convexity		Right sided convexity/right side gets wider and fuller
Lateral strains	2 vertical axes, rotation in same direction	<u>Left Lateral Strain</u> - base of the sphenoid pointing left		5th digits point left
		<u>Right Lateral Strain</u> - base of sphenoid pointing right		5th digits point right
Vertical strains	2 transverse axes, rotation in same directions	<u>Inferior Vertical</u> base of sphenoid points inferiorly		2^{nd} digits move superiorly
		<u>Superior Vertical Strain</u> - base of sphenoid points superiorly		2^{nd} digits move inferiorly

Rib Dysfunctions

B ottom [rib is key rib]
I nhalation [for inhalation SD]
T op [rib is key rib]
E xhalation [for exhalation SD]

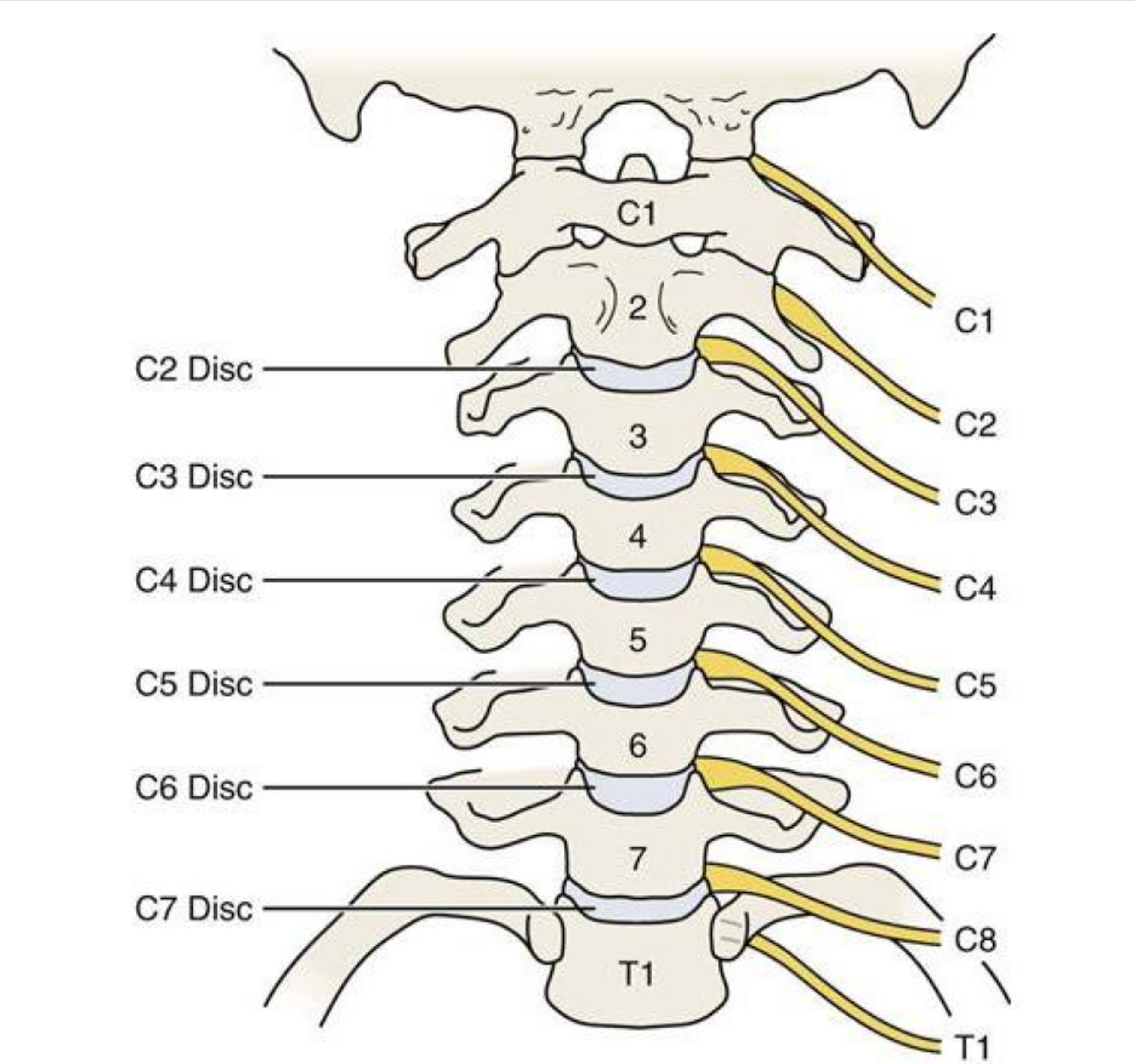
For treatment of exhaled ribs with ME remember:

Ribs	Muscles
Rib 1	Anterior & Middle Scalenes
Rib 2	Posterior Scalene
Ribs 3-5	Pectoralis Minor
Ribs 6-8	Serratus Anterior
Ribs 9-11	Latissimus Dorsi
Rib 12	Quadratus Lumborum



There are 7 cervical vertebrae and 8 cervical nerve roots.

In the cervical spine the nerve exits above its corresponding vertebrae.



https://clinicalgate.com/cervical-spine-3/

Contraindications to OMT

Absolute Contraindications to HVLA
Osteoporosis
Osteomyelitis, fractures, or bone mets in the area of thrust
Joint instability (i.e. patients with severe RA, or Down syndrome, the transverse ligament of the dens may weaken, resulting in AA subluxation)
Severe herniated nucleus propulsus with radiculopathy
Joint replacement in the area of thrust
Vertebrobasilar insufficiency
Joint (including vertebral) ankylosis

Relative Contraindications to HVLA
Mild to moderate strain in the area of thrust (i.e. acute whiplash)
Pregnancy
Post-surgical conditions
Mild to moderae herniated nucleus propulsus with radiculopathy
Patients on anticoagulant therapy or hemophiliacs should be treated with caution to prevent bleeding
Osteoarthritic joints with moderate motion loss

TART

Findings	Acute	Chronic
Tissue texture changes Muscle Skin Soft tissue	Increased tone, spasm Warm, moist, red, inflammed Boggy, edematous, fluid build up from vascular leakage	Decreased tone, flaccid, mushy Cool, pale Doughy, stringy, fibrotic, thickened, contracted
Asymmetry	Present	Present with compensation in other areas of the body
Restriction	Present, painful with movement	Present, decreased or no pain
Tenderness	Greatest	Present but less
Viscero-somatic reflexes	Uncommon or minimal	Common

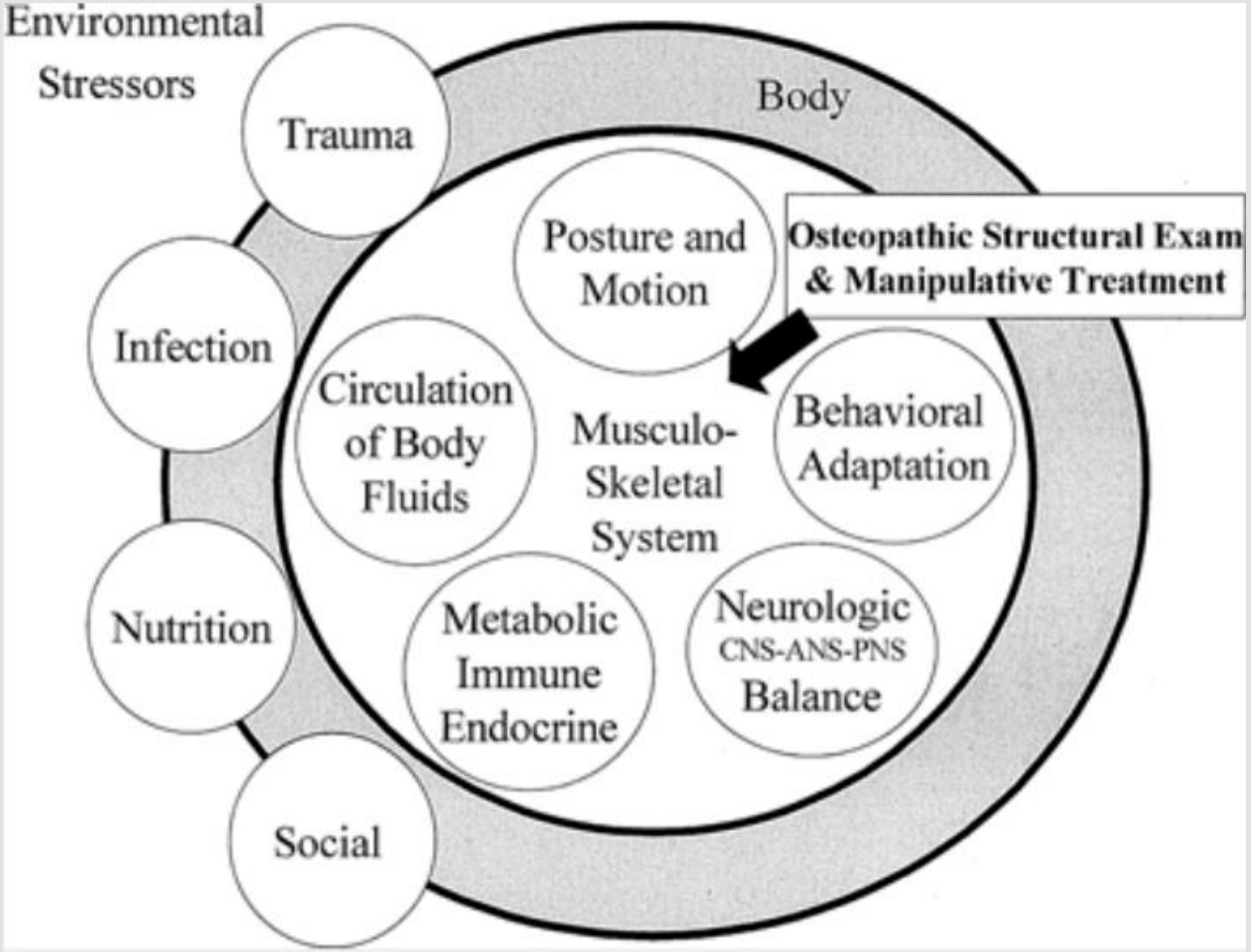
Important note - Classically, prolonged blanching with deep tissue texture assessment in lighter-skinned individuals is associated with chronic changes. In darker-skinned individuals, prolonged blanching can be associated with both acute and chronic changes.

The Five Osteopathic Models

1. Biomechanical model
2. Respiratory-circulatory model
3. Neurological model
4. Metabolic-energy model
5. Behavioral model

ABC'S of Osteopathy (Touro)

- Autonomics
- Biomechanics
- Circulation
- Screening



Foundations of Osteopathic Medicine

Trigger point vs. Counterstrain points

TRIGGER POINT	COUNTERSTRAIN POINT
Patient presents with characteristic pain pattern	No associated pain pattern
Located in muscle tissue	Located in muscle, tendons, ligaments and fascia
Locally tender	Locally tender
Patient jumps when pressed	Patient jumps when pressed
Elicits a radiating pain pattern when pressed	No radiating pattern when pressed
Present within a taut band of tissue	Taut band not present
Elicits twitch response with plucking motion	Twitch response not present
Dermographia of skin over point	Dermographia not present
Travell felt pathology was the trigger point	Jones felt Counterstrain point was manifestation of a somatic dysfunction
Point treated by stretching and spraying muscle containing point or injection	Point treated by positioning that eliminates tenderness

Also can be found in Intro to Counterstrain lecture (week 6 of OPP2)

TUCOM OMM Manual

Supination – Radial head glides anteriorly

Pronation – Radial head glides posteriorly

P with P

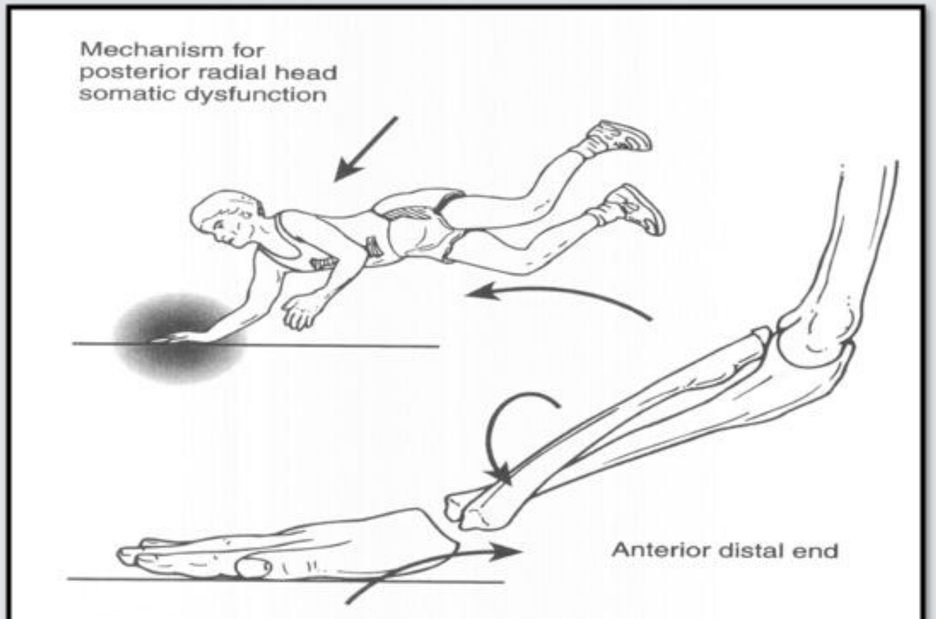
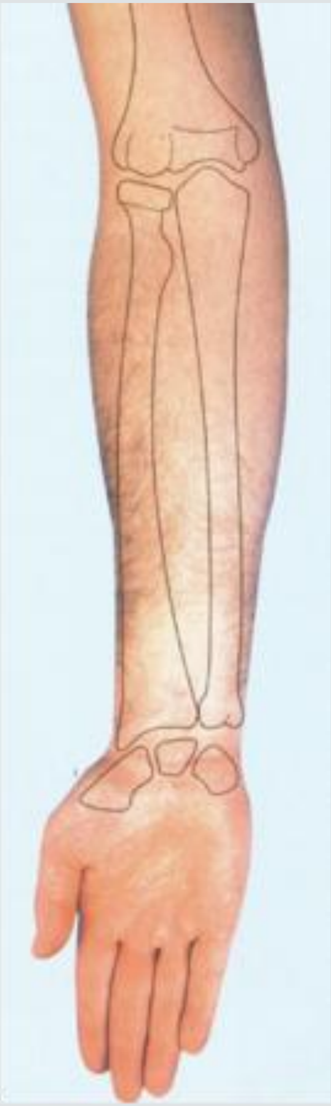


FIGURE 47.9. Forward fall on outstretched hand. (Illustration by W. A. Kuchera.)

Pronation



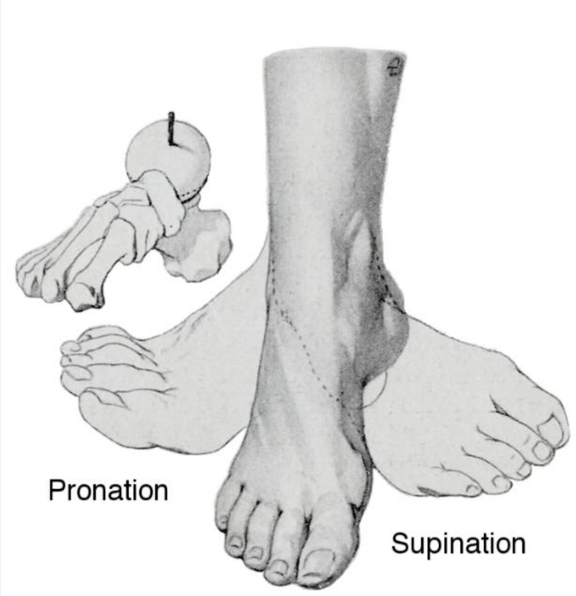
Supination

Inversion Ankle Sprain

– ***Grade 1 Sprain:*** *slight stretching and damage to fibers (fibrils) of ligament*
- Mild joint pain reproduced on local palpation
- Firm endpoint with minimal laxity, no functional loss
- No joint instability
- Tear or injury of Anterior Talofibular ligament

ME for Posterior Fibular Head SD

- Fibular head glides *posteriorly* with **supination** of foot.
- Fibular head glides *posteriorly* with **internal rotation** of tibia.
- Fibular head glides *anteriorly* with **pronation** of foot.
- Fibular head glides *anteriorly* with *external rotation* of tibia.



Supination:
- Inversion of hindfoot
- Adduction of forefoot
- Plantarflexion of talocrural regions

Dr. Moloney's Clinical Considerations of the Knee and Ankle,
TUCOM OMM Manual
https://en.wikipedia.org/wiki/Pronation_of_the_foot

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